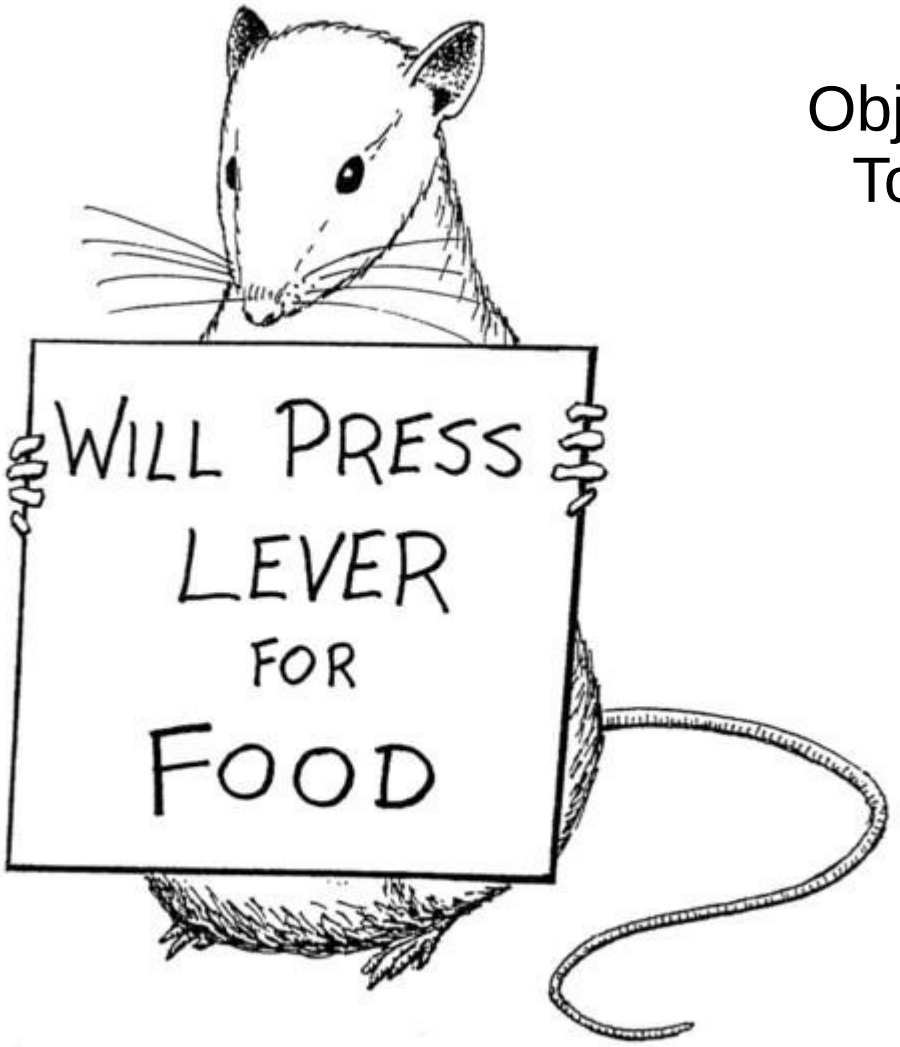
Practical RL

Episode 3; 2024

Model-free reinforcement learning

Recap: discounted rewards



Objective:
Total action value

$$G_t = r_t + \gamma \cdot r_{t+1} + \gamma^2 \cdot r_{t+2} + \ldots + \gamma^n \cdot r_{t+n}$$

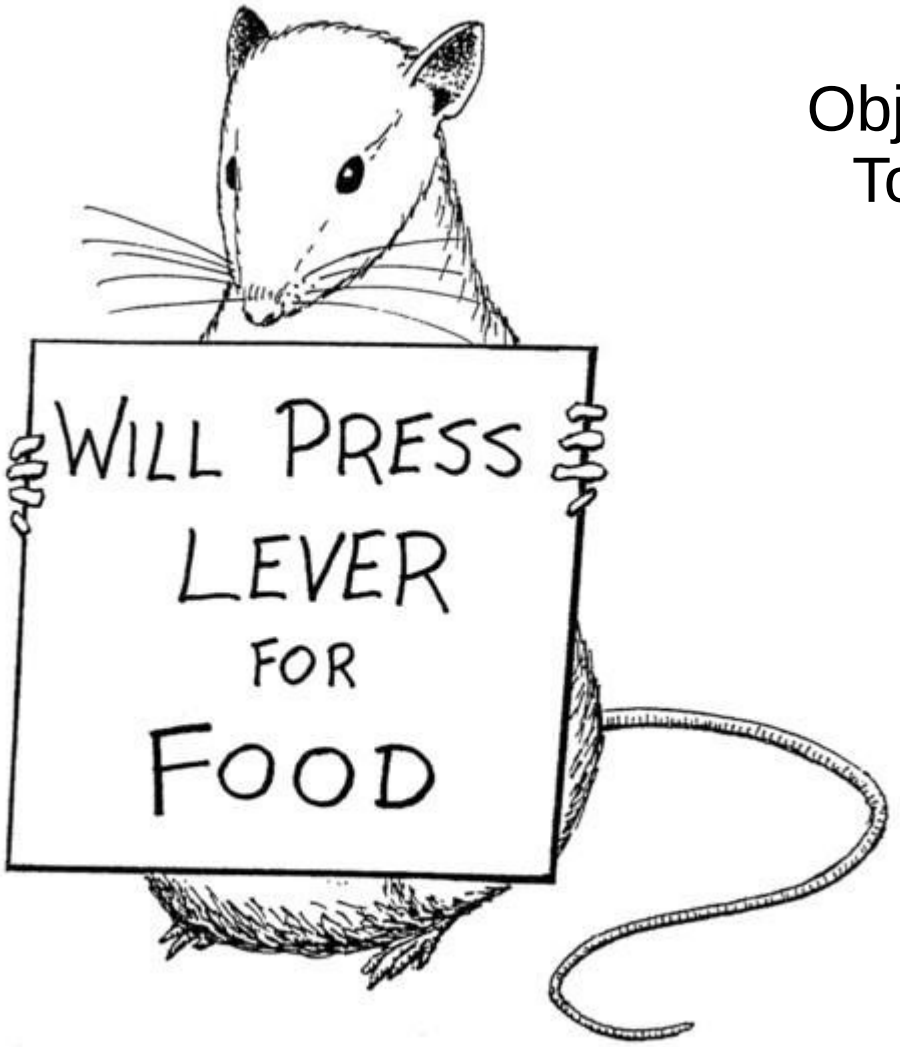
$$G_t = \sum_i \gamma^i \cdot r_{t+i} \qquad \gamma \in (0,1) const$$

γ ~ patience
Cake tomorrow is γ as good as now

Reinforcement learning:
- Find policy that maximizes the expected reward

$$\pi = P(a|s) : E[G] \rightarrow max$$

Recap: discounted rewards



Objective:
Total action value

$$G_t = r_t + \gamma \cdot r_{t+1} + \gamma^2 \cdot r_{t+2} + \ldots + \gamma^n \cdot r_{t+n}$$

$$G_t = \sum_i \gamma^i \cdot r_{t+i} \qquad \gamma \in (0,1) \; const$$

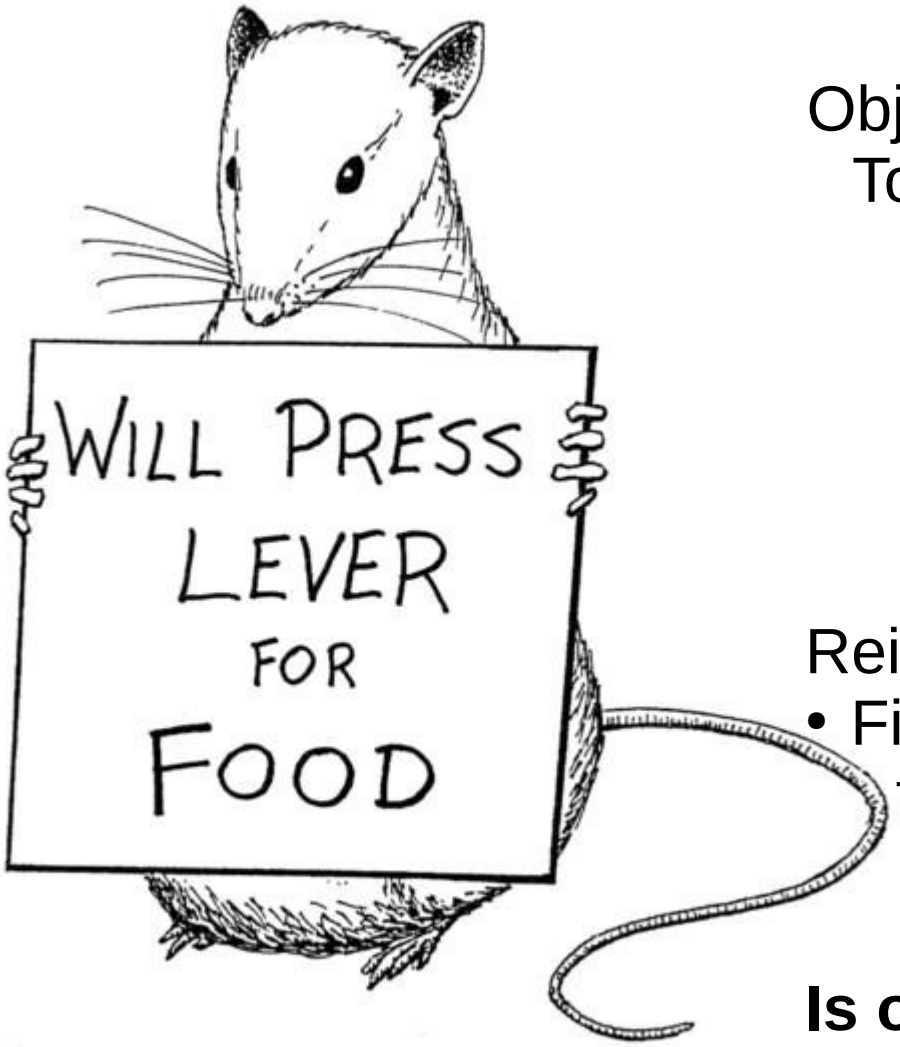
Trivia: which γ corresponds to "only current reward matters"?

Reinforcement learning:

- Find policy that maximizes the expected reward

$$\pi = P(a|s) : E[G] \rightarrow max$$

Recap: discounted rewards



Objective:
Total action value

$$G_t = r_t + \gamma \cdot r_{t+1} + \gamma^2 \cdot r_{t+2} + \ldots + \gamma^n \cdot r_{t+n}$$

$$G_t = \sum_i \gamma^i \cdot r_{t+i} \qquad \gamma \in (0,1) const$$

Reinforcement learning:

- Find policy that maximizes the expected reward

$$\pi = P(a|s) : E[G] \rightarrow max$$

Is optimal policy same as it would be in monte-carlo (if we add-up all r_t)?

Previously...

- **V(s)** and **V*(s,a)**

WTF is V(s)?

Previously...

- **V(s)** and **V*(s,a) –** state values
- know V* and P(s'|s,a) → know optimal policy
- We can learn V* with dynamic programming

$$V_{i+1}(s) := max_a [r(s,a) + \gamma \cdot E_{s' \sim P(s'|s,a)} V_i(s')]$$

Recap: notation

- **$V_\pi(s)$** – expected G from state **s** if you follow **π**
- **V*(s)** – expected G from state s if you follow **π***

* where $G_t = r_t + \gamma \cdot r_{t+1} + \gamma^2 \cdot r_{t+2} + ... + \gamma^n \cdot r_{t+n}$

Recap: notation

- $\mathbf{V_\pi(s)}$ – expected G from state **s** if you follow $\boldsymbol{\pi}$
- $\mathbf{V^*(s)}$ – expected G from state s if you follow $\boldsymbol{\pi^*}$

- $\mathbf{Q_\pi(s,a)}$ – expected G from state **s**
  - if you start by taking action **a**
  - and follow $\boldsymbol{\pi}$ from next state on

- $\mathbf{Q^*(s,a)}$ – guess what it is :)

Recap: notation

- **$V_\pi(s)$** – expected G from state **s** if you follow **π**
- **$V^*(s)$** – expected G from state s if you follow **π***

- **$Q_\pi(s,a)$** – expected G from state **s**
  - if you start by taking action **a**
  - and follow **π** from next state on

- **$Q^*(s,a)$** – same as **$Q_\pi(s,a)$** where **$\pi = \pi^*$**

Trivia

- Assuming you know **Q*(s,a)**,
  - how do you compute **π***
  - how do you compute **V*(s)**?
- Assuming you know V(s)
  - how do you compute **Q(s,a)?**

To sum up

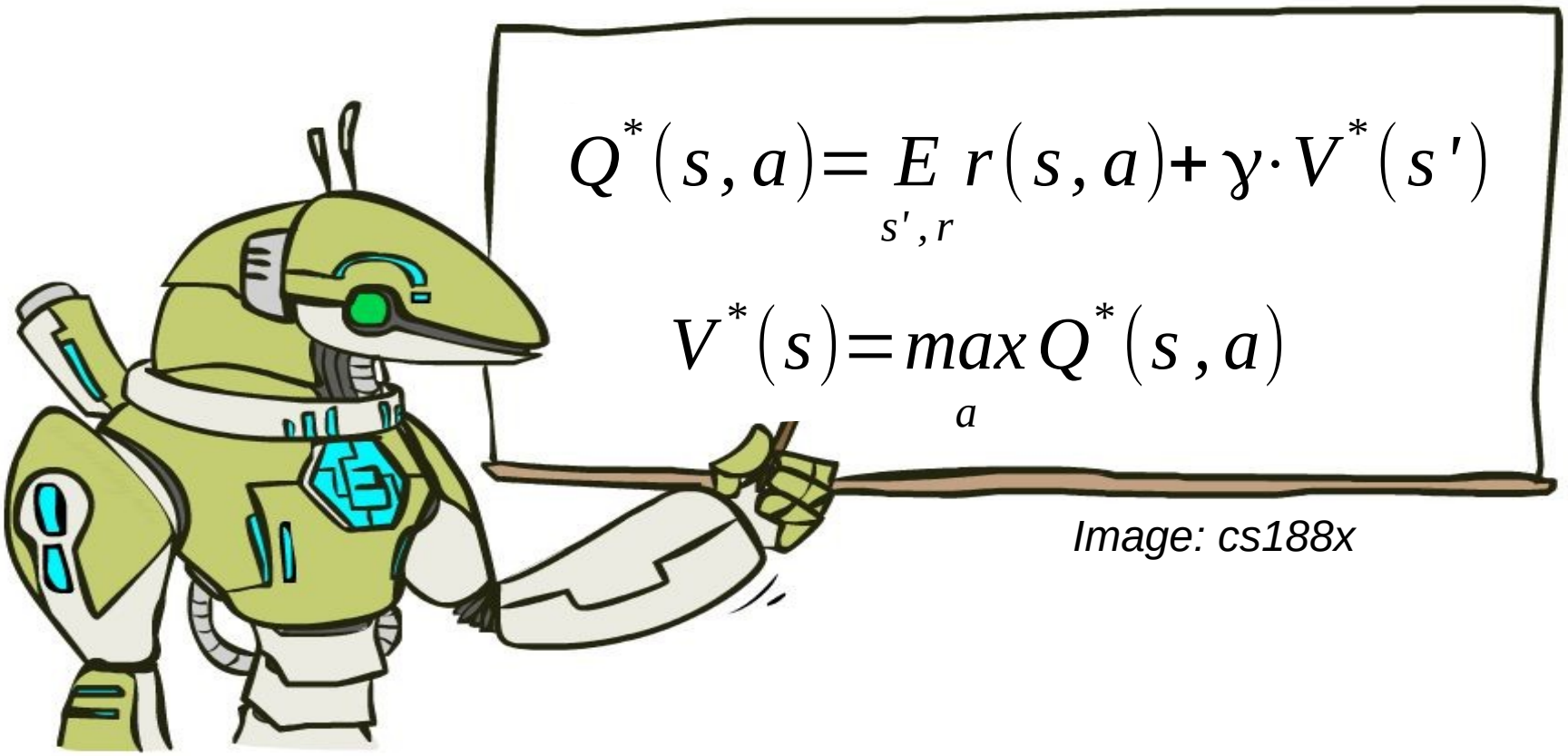
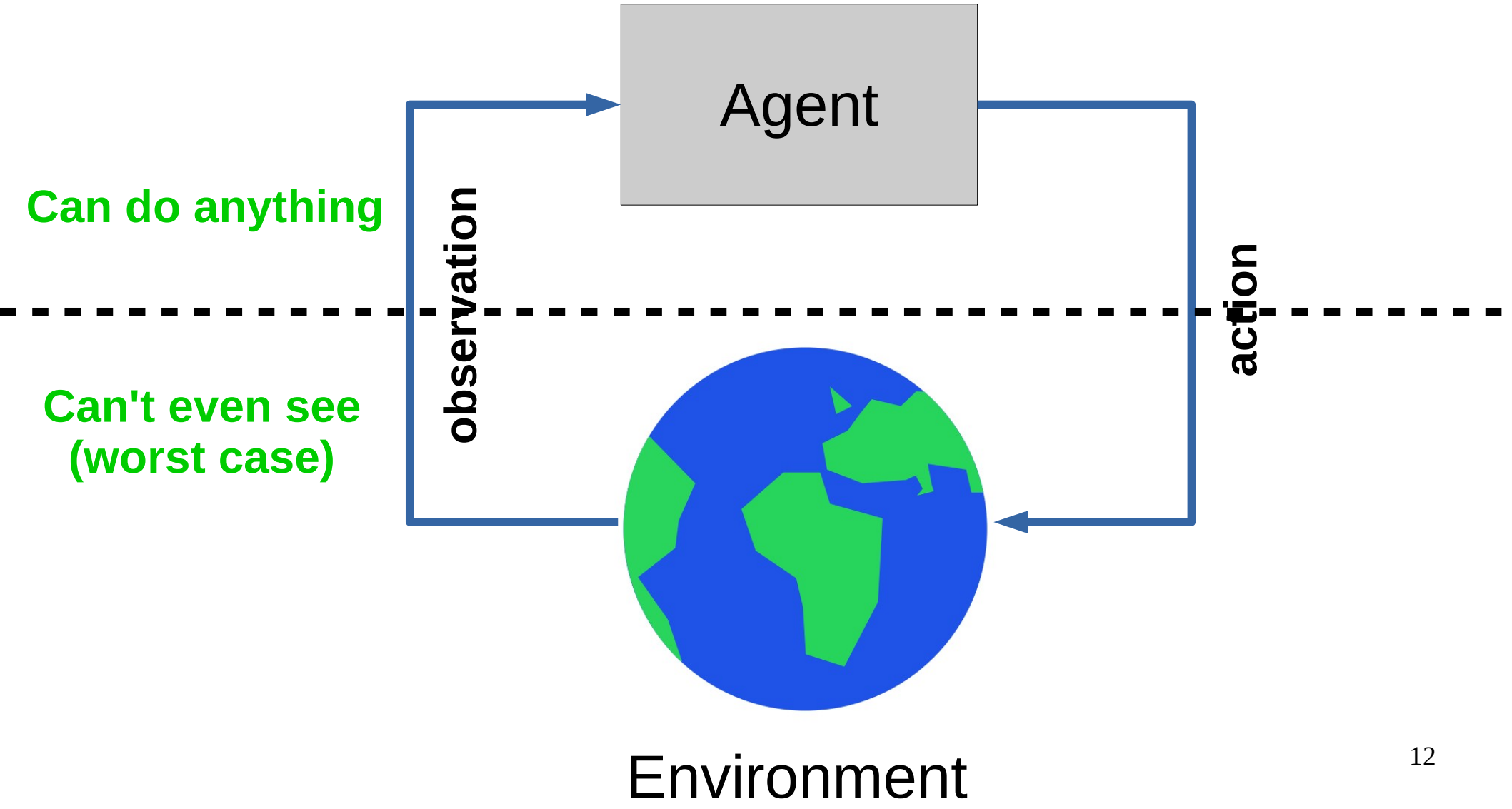


Image: cs188x

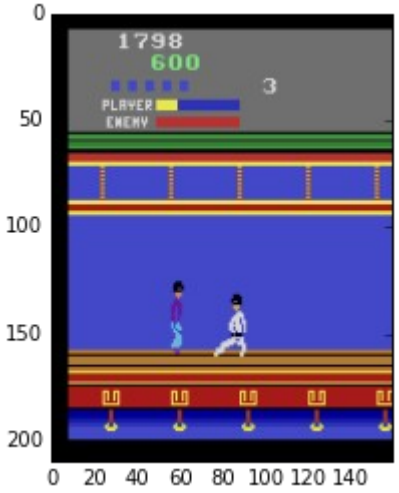
Action value $Q_\pi(s,a)$ is the expected total reward **G** agent gets from state **s** by taking action **a** and following policy **π** from next state.

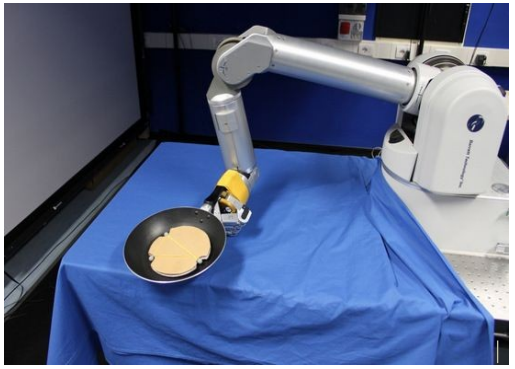
$$\pi(s) : argmax_a Q(s,a)$$

Decision process in the wild



Decision process in the wild





Model-free reinforcement learning:

We don't know actual

$P(s',r|s,a)$

Whachagonnado?

Model-free reinforcement learning:

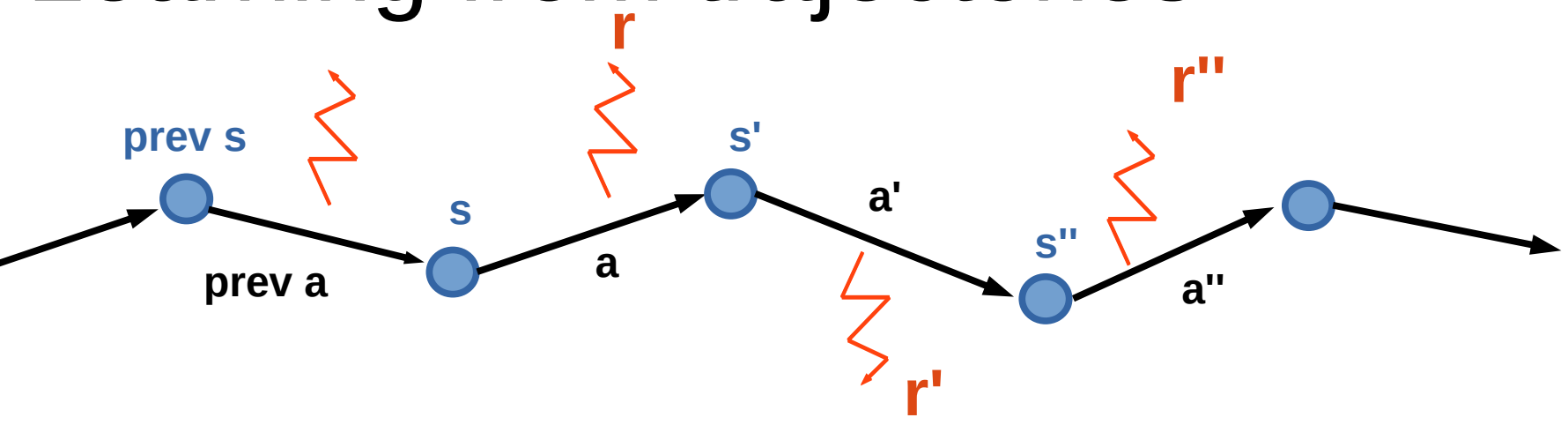
We don't know actual

$P(s',r|s,a)$

Learn it?

Get rid of it?

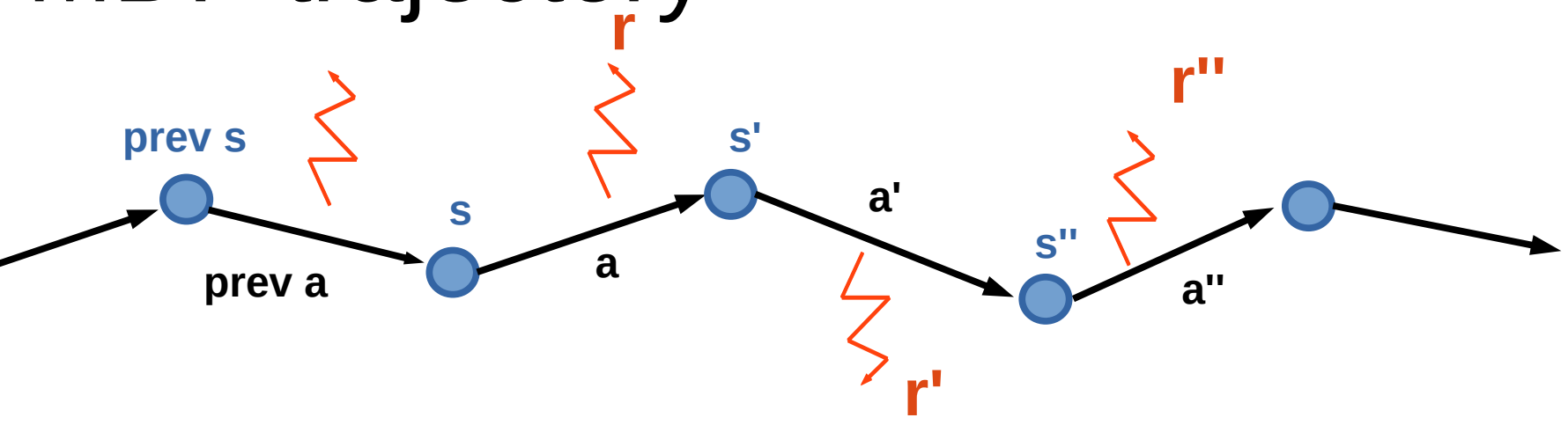
Learning from trajectories



Model-based: you know P(s'|s,a)
- can apply dynamic programming
- can plan ahead

Model-free: you can sample trajectories
- can try stuff out
- insurance not included

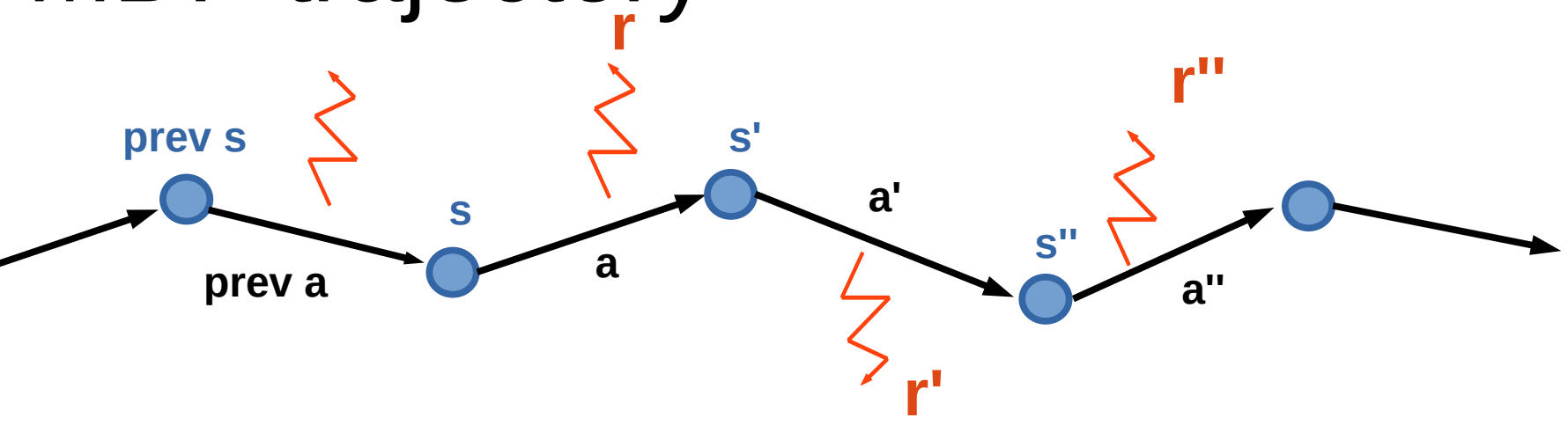
MDP trajectory



- Trajectory is a sequence of
  - states (s)
  - actions (a)
  - rewards (r)

- We can only sample trajectories

MDP trajectory

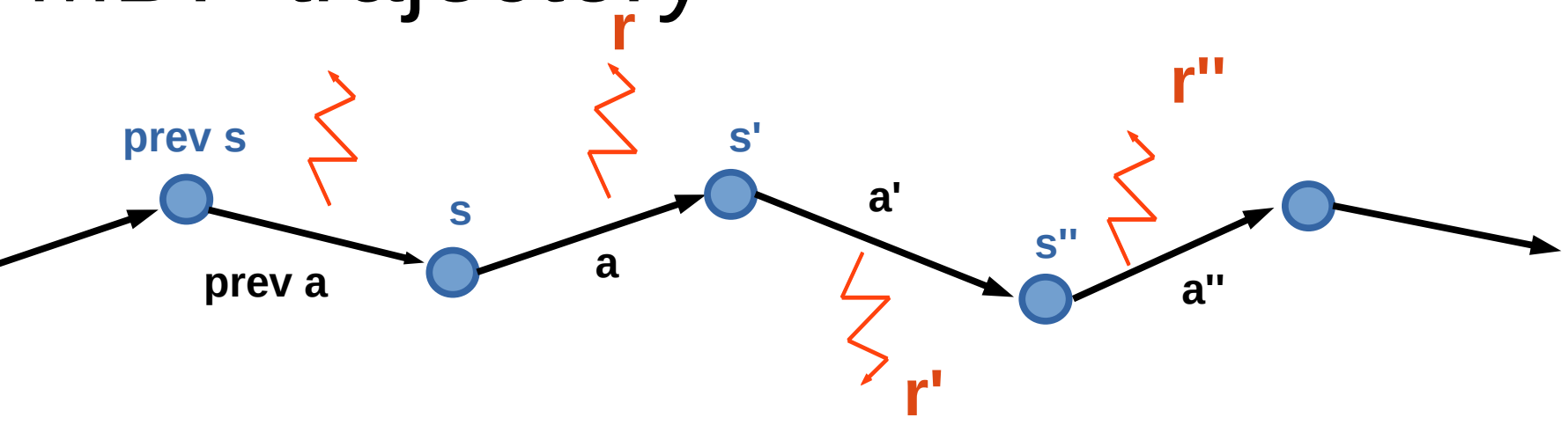


- Trajectory is a sequence of
  - states (s)
  - actions (a)
  - rewards (r)

Q: What to learn?
V(s) or Q(s,a)

- We can only sample trajectories

MDP trajectory



- Trajectory is a sequence of
  - states (s)
  - actions (a)
  - rewards (r)

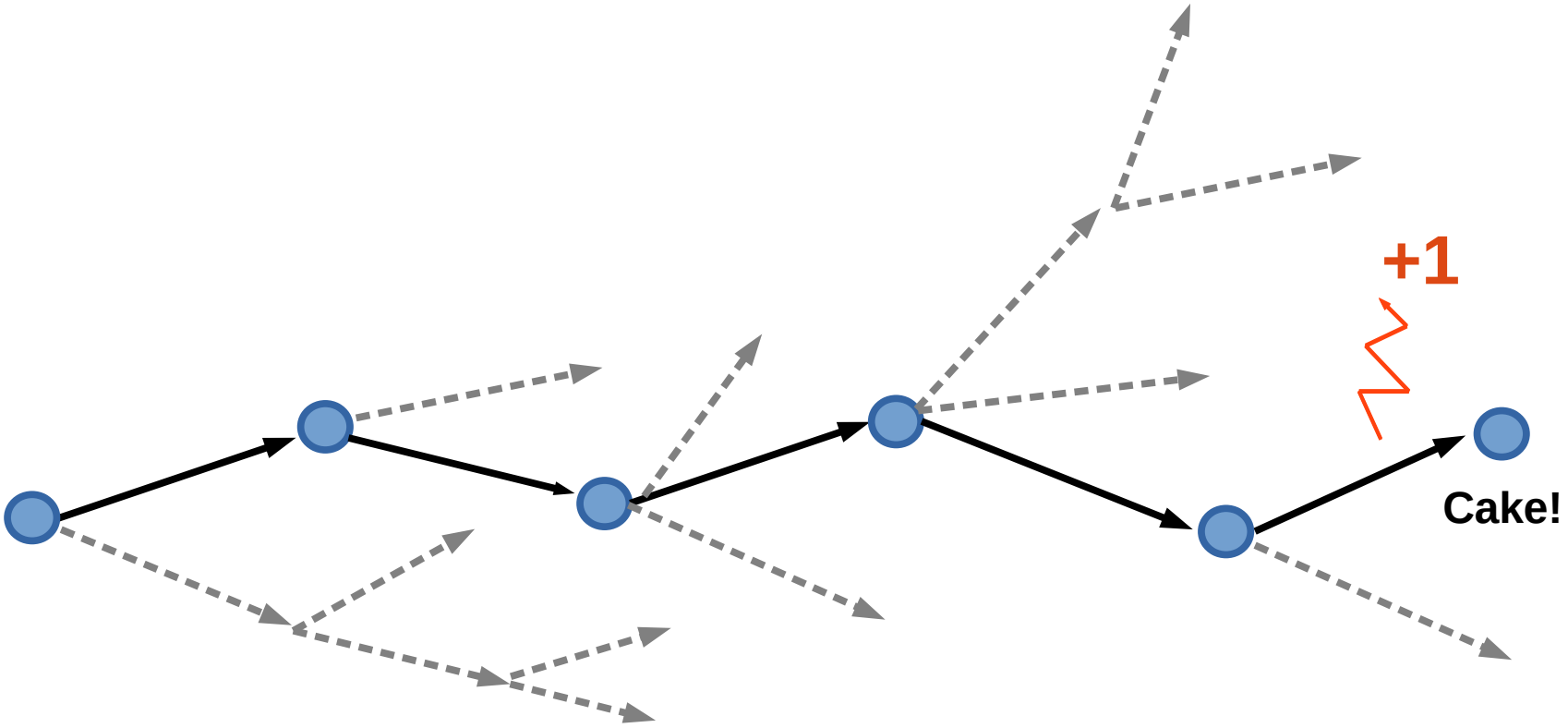
Q: What to learn?
V(s) or Q(s,a)

V(s) is useless
without P(s'|s,a)

- We can only sample trajectories

Idea 1: monte-carlo

- Get all trajectories containing particular (s,a)
- Estimate G(s,a) for each trajectory
- Average them to get expectation



Idea 1: monte-carlo

- Get all trajectories containing particular (s,a)
- Estimate G(s,a) for each trajectory
- Average them to get expectation

takes a lot of sessions

Image: super meat boy

Idea 2: temporal difference

- Remember we can improve Q(s,a) iteratively!

$$Q(s_t, a_t) \leftarrow \underset{r_t, s_{t+1}}{E}\ r_t + \gamma \cdot max_{a'} Q(s_{t+1}, a')$$

Idea 2: temporal difference

- Remember we can improve Q(s,a) iteratively!

$$Q(s_t, a_t) \leftarrow \underset{r_t, s_{t+1}}{E}\ r_t + \gamma \cdot max_{a'} Q(s_{t+1}, a')$$

That's Q*(s,a)

That's value for π*
aka optimal policy

Idea 2: temporal difference

- Remember we can improve Q(s,a) iteratively!

$$Q(s_t, a_t) \leftarrow \underset{r_t, s_{t+1}}{E}\ r_t + \gamma \cdot max_{a'} Q(s_{t+1}, a')$$

That's Q*(s,a)

That's value for π*
aka optimal policy

That's something we don't have

What do we do?

Idea 2: temporal difference

Idea 2: temporal difference

- Replace expectation with sampling

$$\underset{r_t, s_{t+1}}{E}\left[r_t + \gamma \cdot max_{a'} Q(s_{t+1}, a') \right] \approx \frac{1}{N} \sum_i r_i + \gamma \cdot max_{a'} Q\left(s_i^{next}, a'\right)$$

Idea 2: temporal difference

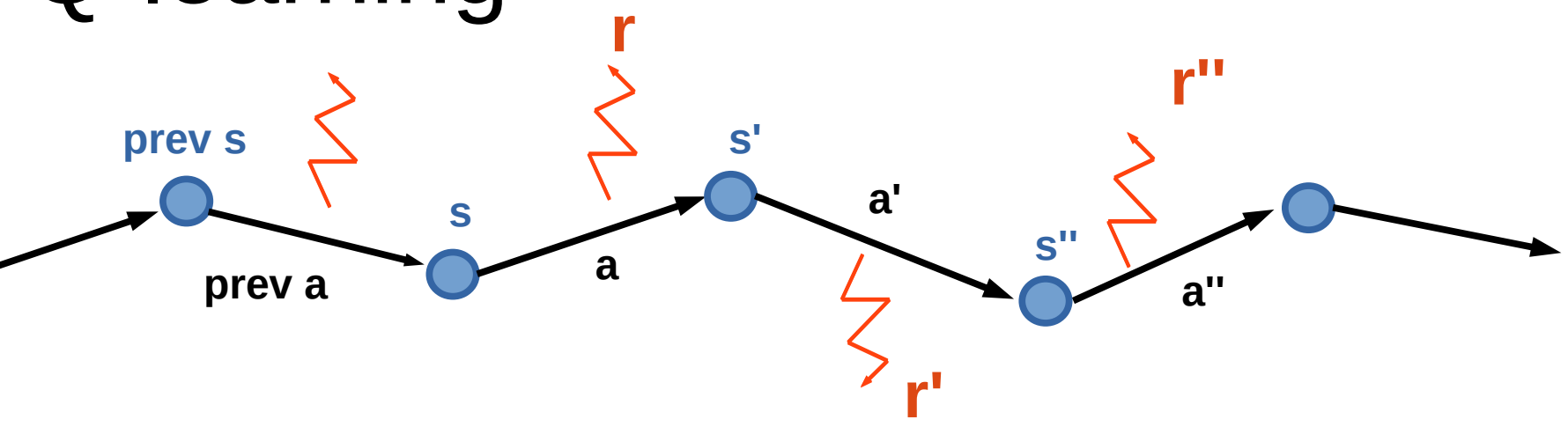
- Replace expectation with sampling

$$\underset{r_t, s_{t+1}}{E}\left[r_t + \gamma \cdot max_{a'} Q(s_{t+1}, a') \right] \approx \frac{1}{N} \sum_i r_i + \gamma \cdot max_{a'} Q\left(s_i^{next}, a'\right)$$

- Use moving average with just one sample!

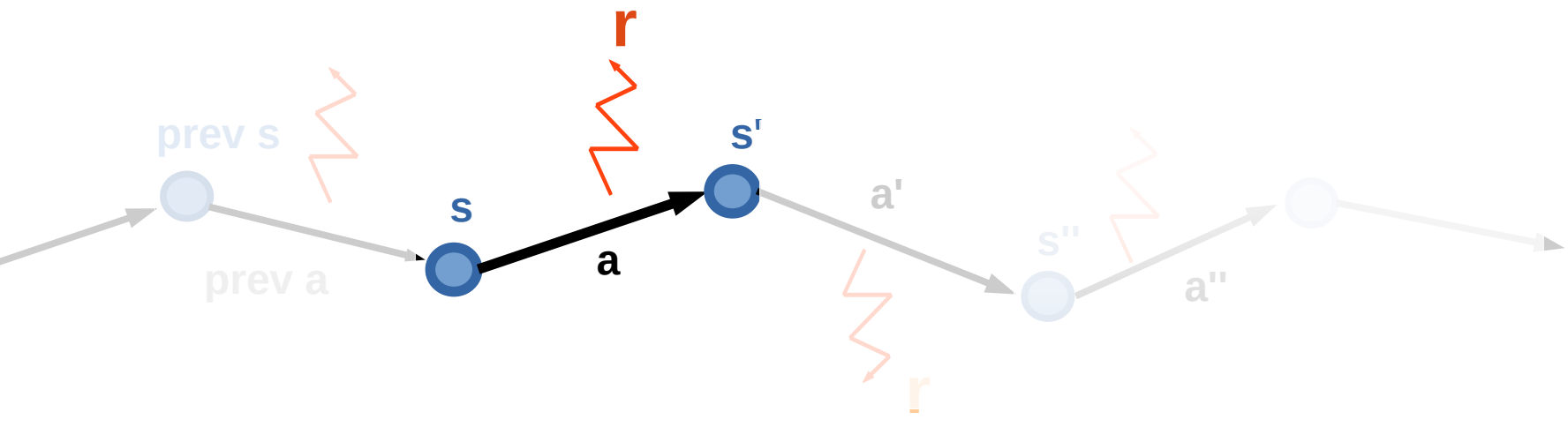
$$Q(s_t, a_t) \leftarrow \alpha \cdot \left(r_t + \gamma \cdot max_{a'} Q(s_{t+1}, a') \right) + (1 - \alpha) Q(s_t, a_t)$$

Q-learning



- Works on a sequence of
  - states (s)
  - actions (a)
  - rewards (r)

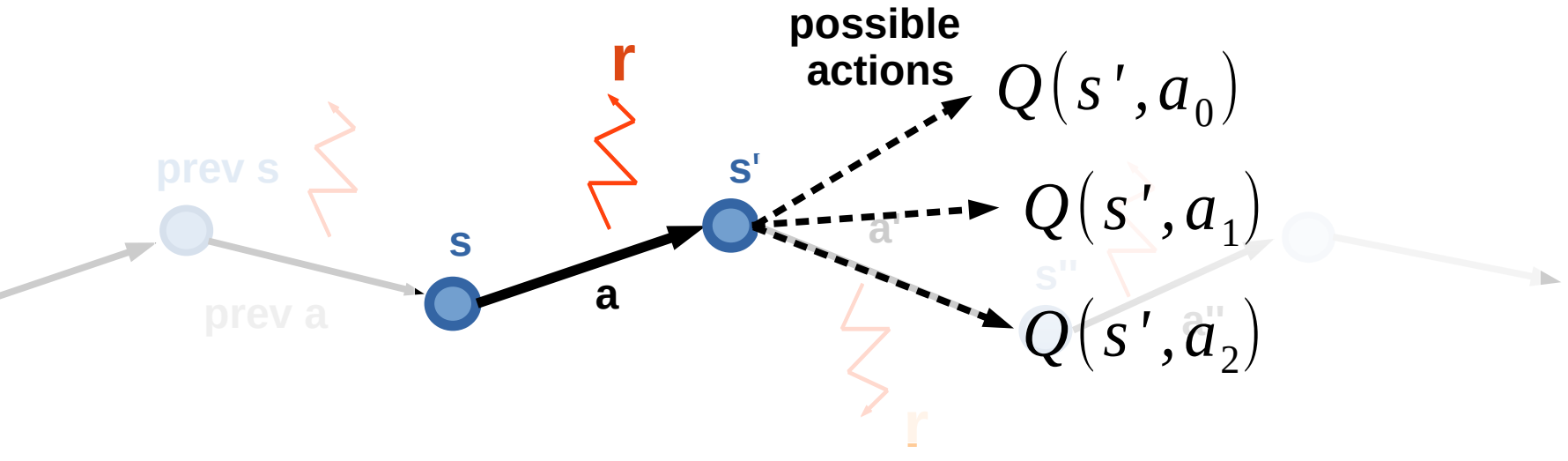
Q-learning



Initialize Q(s,a) with zeros

- Loop:
  - Sample <**s**,**a**,**r**,**s'**> from env

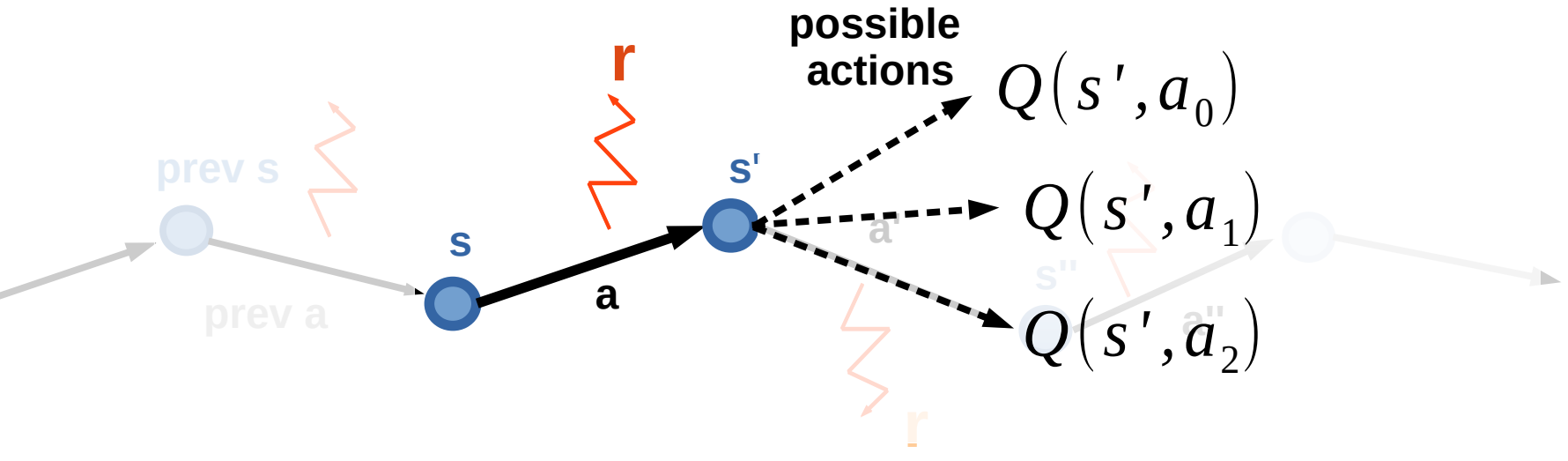
Q-learning



Initialize Q(s,a) with zeros

- Loop:
  - Sample <**s**,**a**,**r**,**s'**> from env
  - Compute $\hat{Q}(s,a) = r(s,a) + \gamma \max_{a_i} Q(s',a_i)$

Q-learning



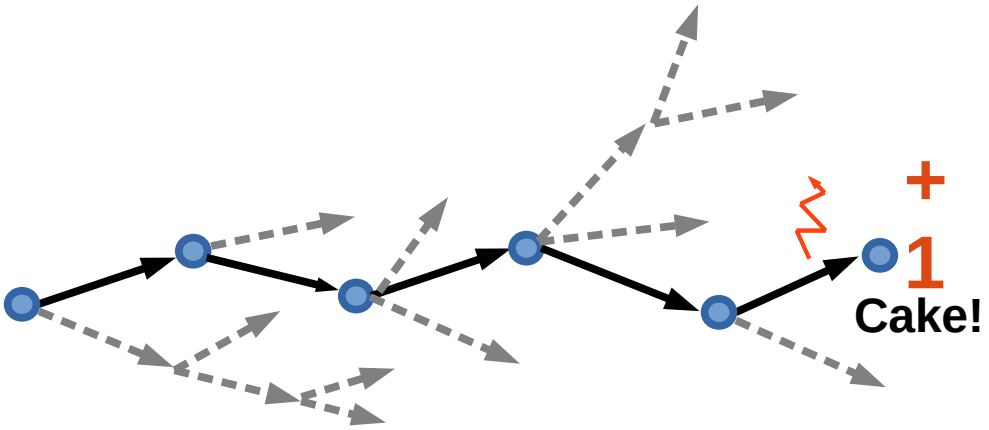
Initialize Q(s,a) with zeros

- Loop:
  - Sample <**s**,**a**,**r**,**s'**> from env
  - Compute $\hat{Q}(s,a) = r(s,a) + \gamma \max_{a_i} Q(s',a_i)$
  - Update $Q(s,a) \leftarrow \alpha \cdot \hat{Q}(s,a) + (1-\alpha) Q(s,a)$

Recap

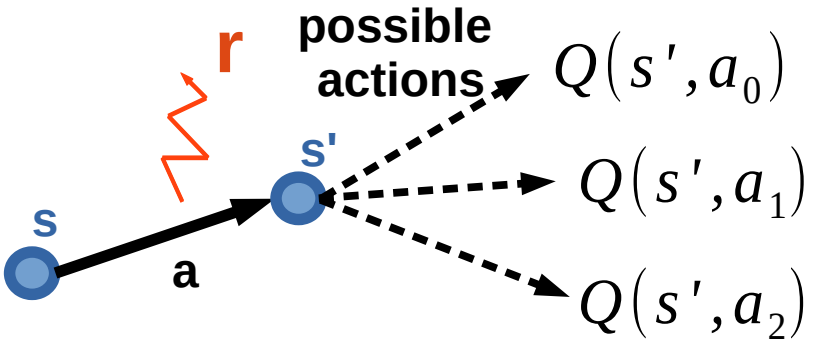
Monte-carlo

- Averages Q over sampled paths



Temporal Difference

- Uses recurrent formula for Q



Nuts and bolts: MC vs TD

Monte-carlo

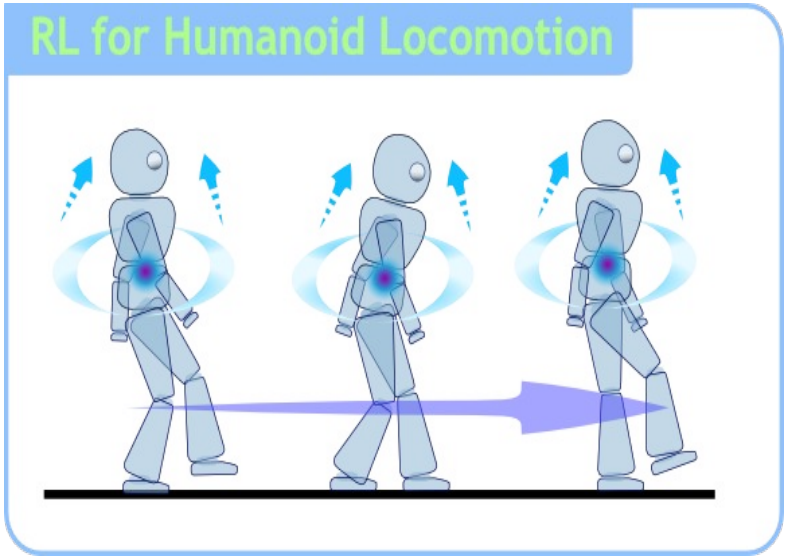
- Averages Q over sampled paths
- Needs full trajectory to learn
- Less reliant on markov property

Temporal Difference

- Uses recurrent formula for Q
- Learns from partial trajectory
  Works with infinite MDP
- Needs less experience to learn

What could possibly go wrong?

Our mobile robot learns to walk.

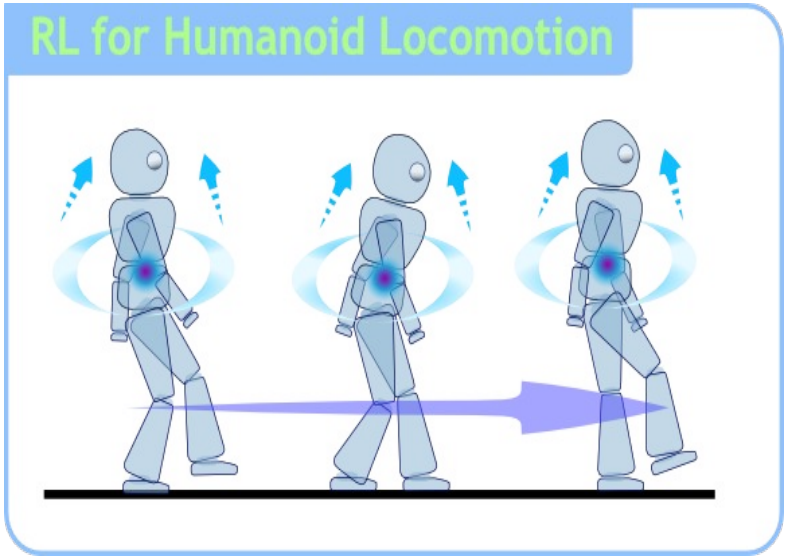


Initial Q(s,a) are zeros
robot uses argmax Q(s,a)

He has just learned to crawl with positive reward!

What could possibly go wrong?

Our mobile robot learns to walk.



Initial Q(s,a) are zeros
robot uses argmax Q(s,a)

Too bad, now he will never learn to walk upright =(

What could possibly go wrong?

New problem:

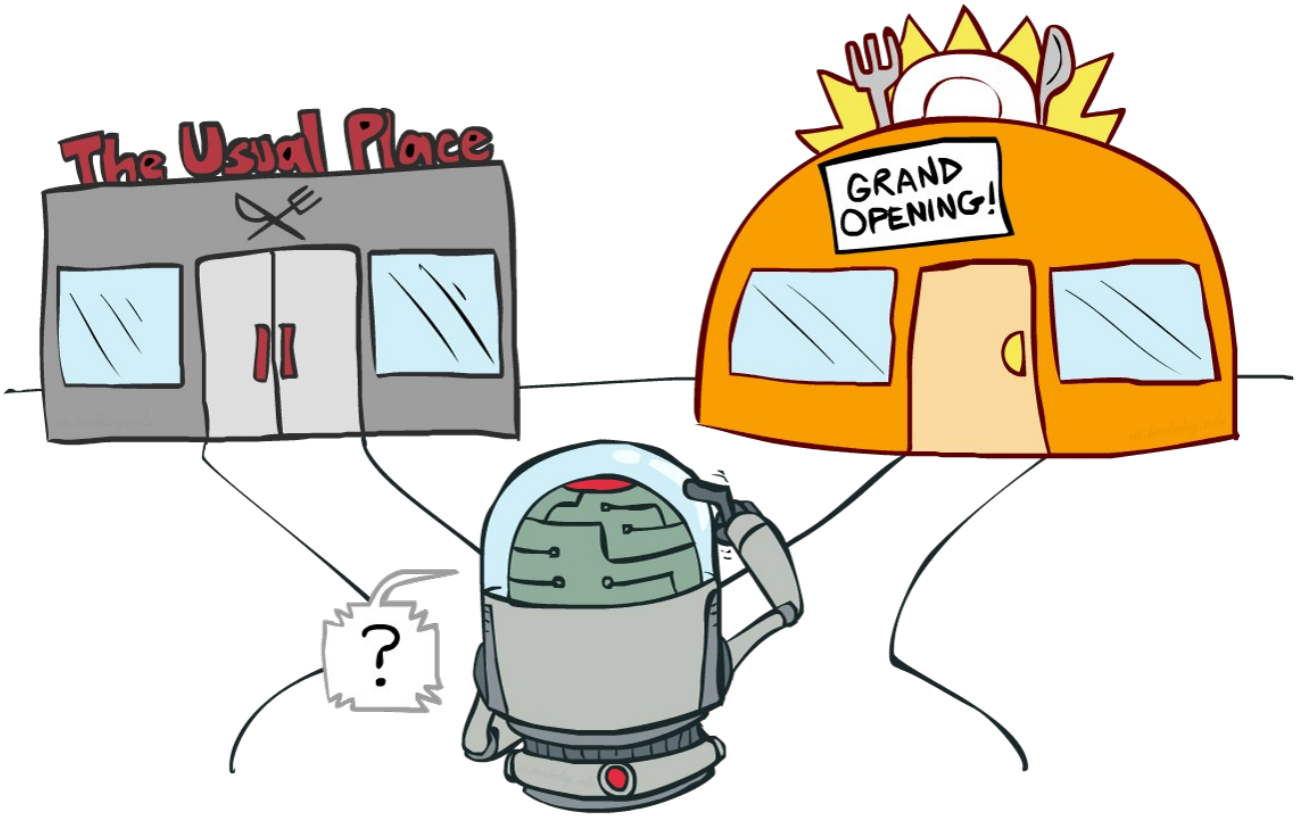
If our agent always takes “best” actions from his current point of view,

How will he ever learn that other actions may be better than his current best one?

Ideas?

Exploration Vs Exploitation

Balance between using what you learned and trying to find something even better



Exploration Vs Exploitation

Strategies:

- ε-greedy
  - With probability ε take random action; otherwise take optimal action.

Exploration Vs Exploitation

Strategies:

- ε-greedy
  - With probability ε take random action; otherwise take optimal action.
- Softmax

  Pick action proportional to softmax of shifted normalized Q-values.

$$\pi(a|s) = softmax\left(\frac{Q(s,a)}{\tau}\right)$$

- More cool stuff coming later

Exploration over time

Idea:

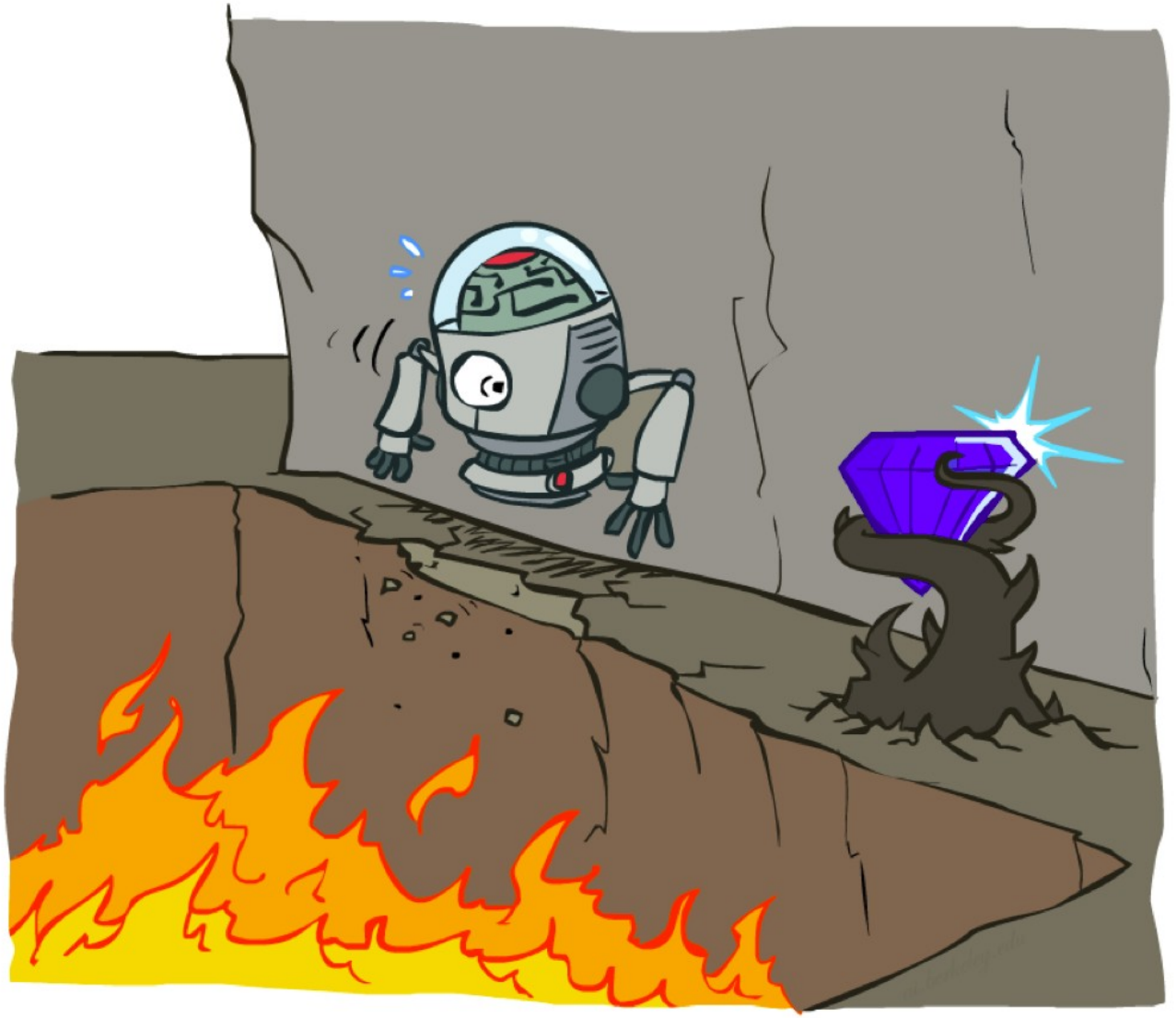
If you want to converge to optimal policy,
you need to gradually reduce exploration

Example:

Initialize ε-greedy ε = 0.5, then gradually reduce it

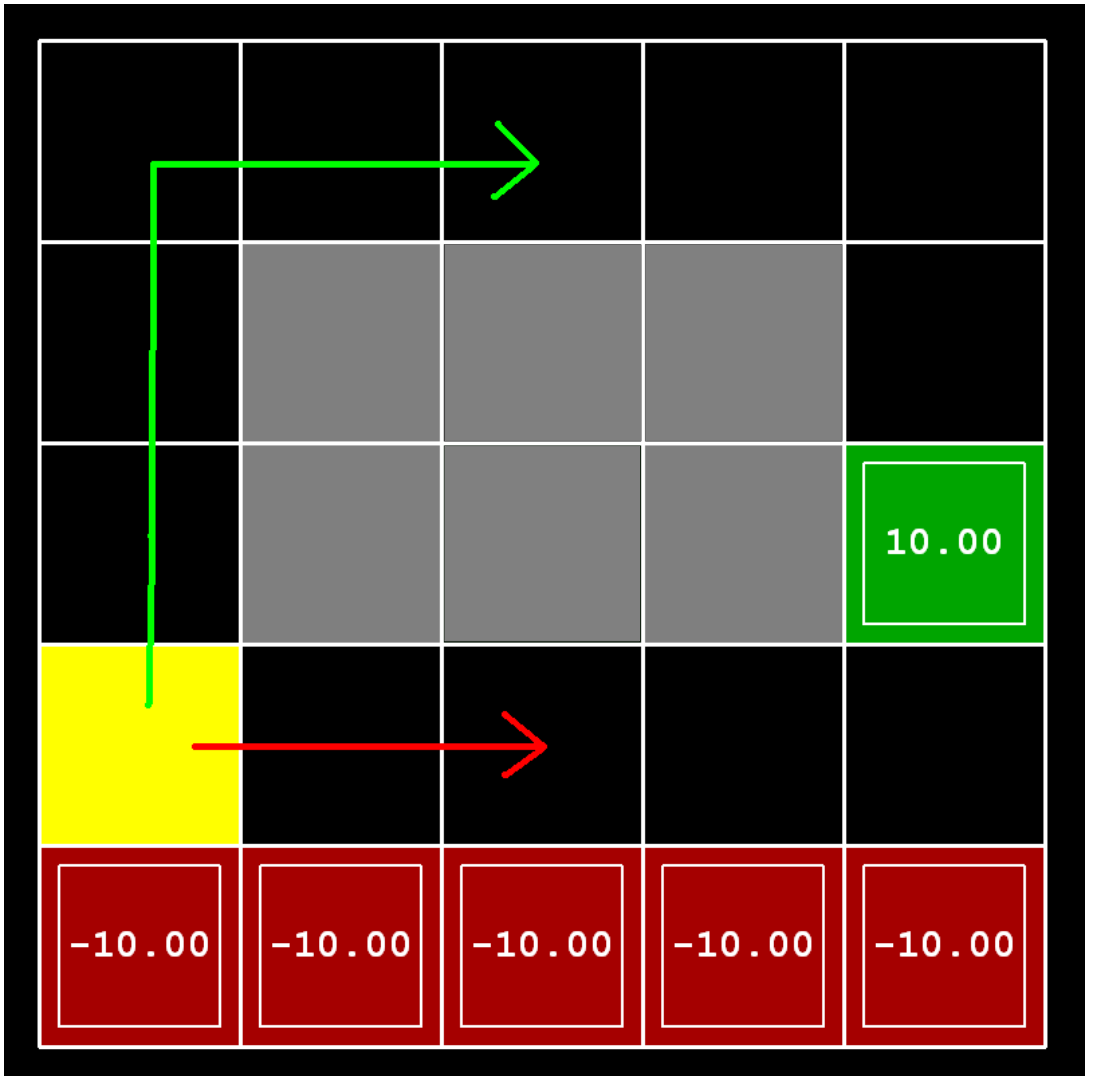
- If $\varepsilon \rightarrow 0$, it's **greedy in the limit**
- Be careful with non-stationary environments

Cliff world



Picture from Berkeley CS188x

Cliff world



Conditions

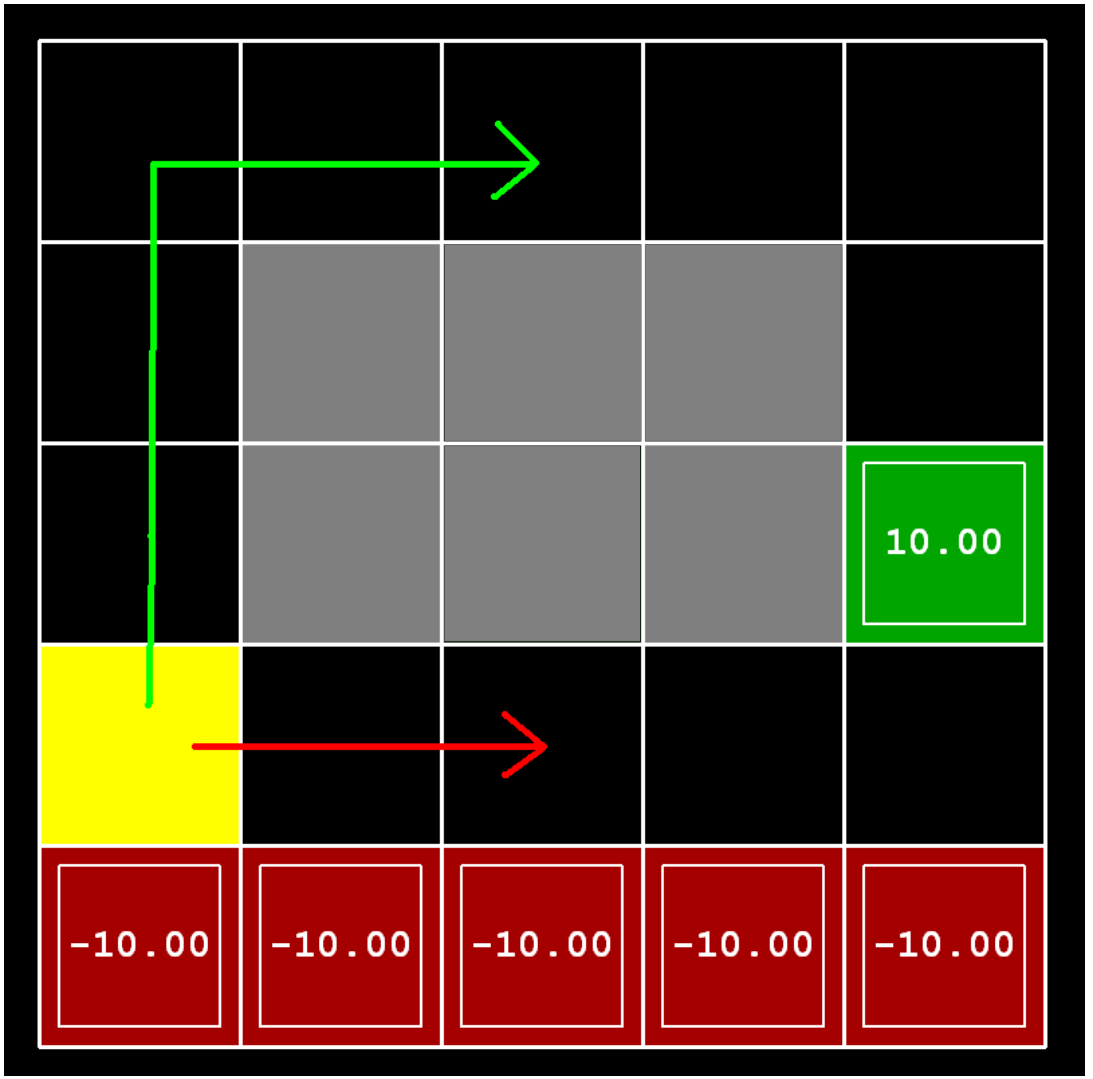
- Q-learning

$\gamma = 0.99 \quad \epsilon = 0.1$

- no slipping

Trivia:
What will q-learning learn?

Cliff world



Conditions

- Q-learning

$\gamma = 0.99 \quad \epsilon = 0.1$

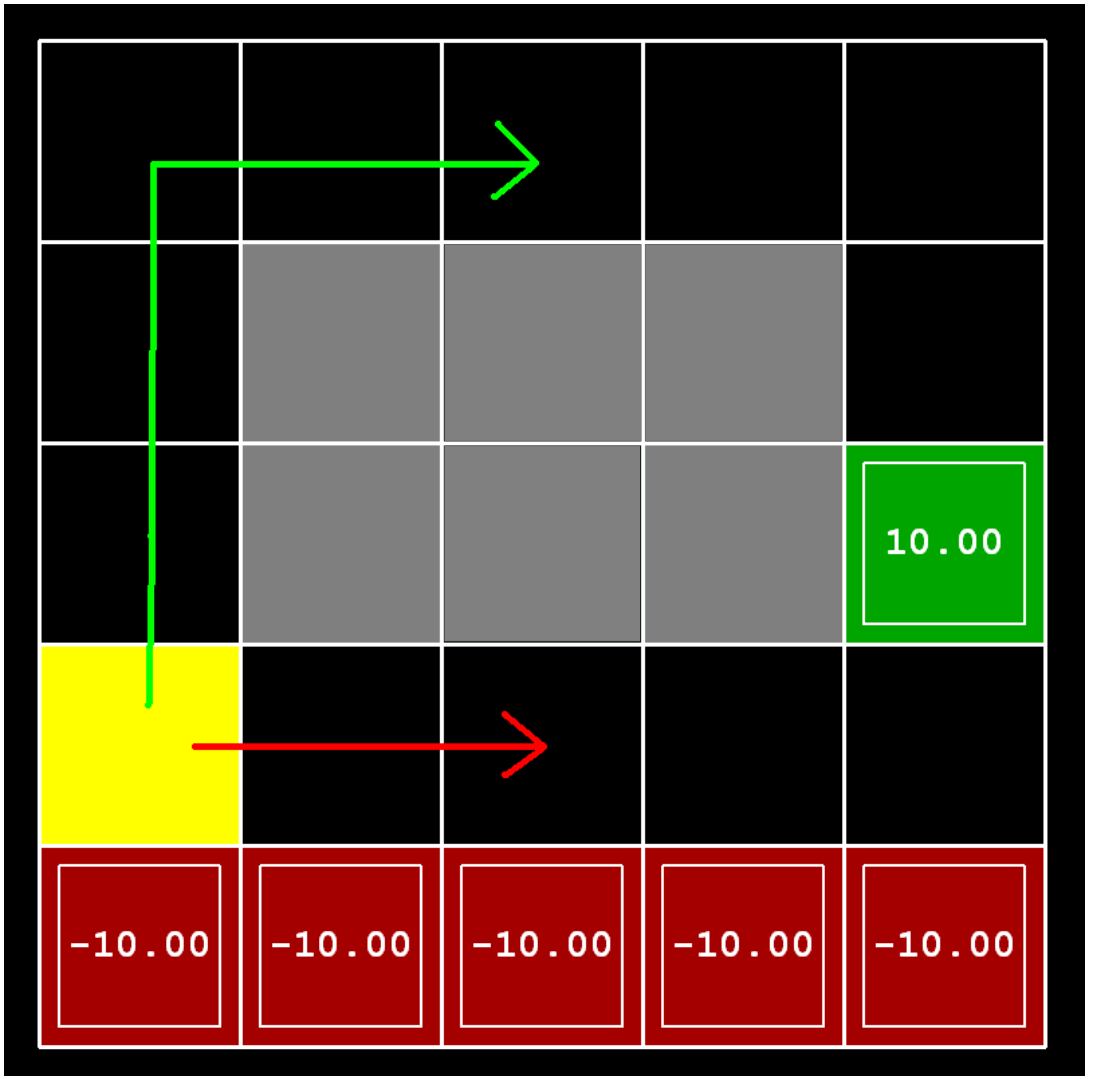
- no slipping

Trivia:
What will q-learning learn?

follow the short path

Will it maximize reward?

Cliff world



Conditions

- Q-learning

$\gamma = 0.99 \quad \epsilon = 0.1$

- no slipping

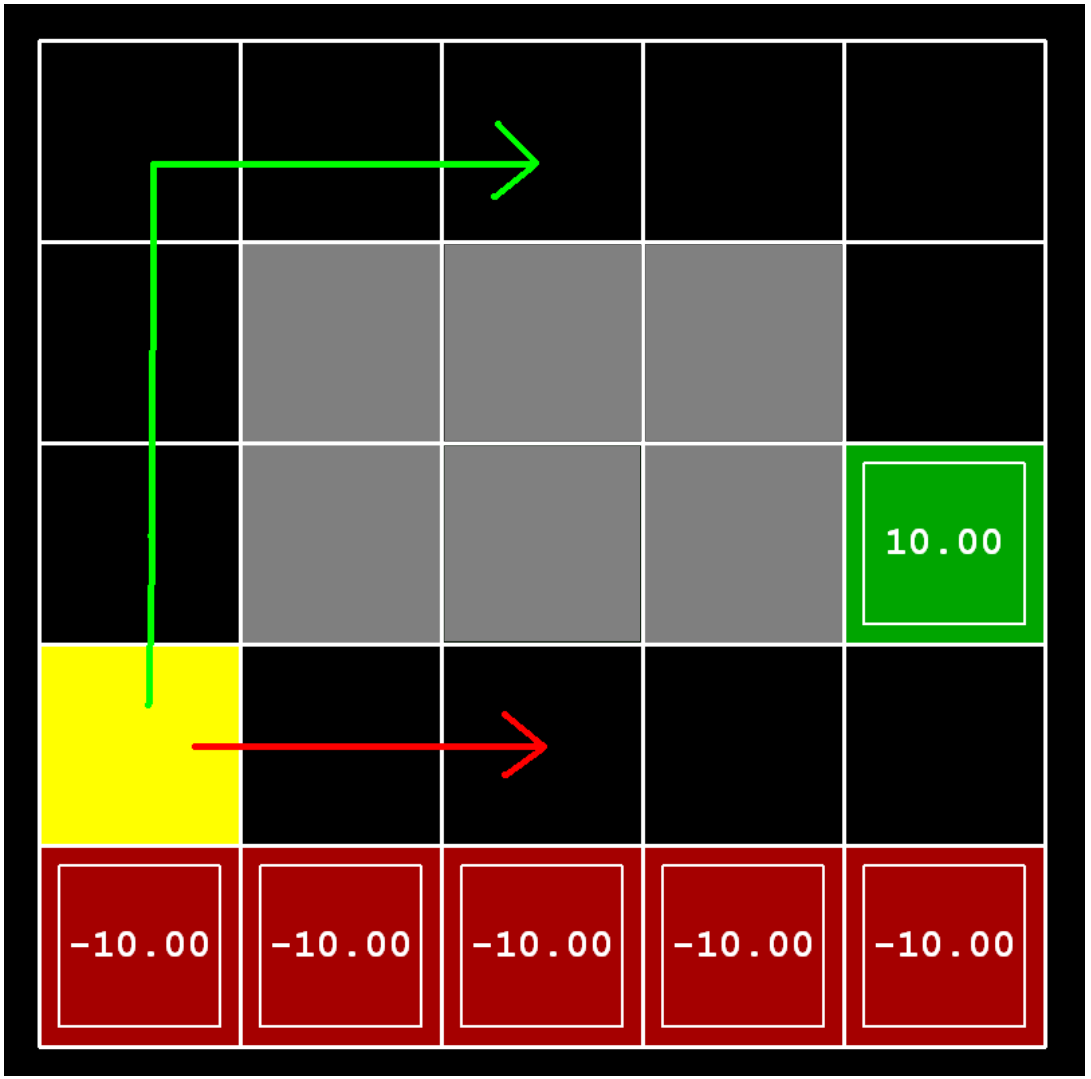
Trivia:
What will q-learning learn?

follow the short path

Will it maximize reward?

no, robot will fall due to epsilon-greedy "exploration"

Cliff world



Conditions

- Q-learning

$\gamma = 0.99 \quad \epsilon = 0.1$

- no slipping

Decisions must account for actual policy!
e.g. ε-greedy policy

Generalized update rule

Update rule (from Bellman eq.)

$$Q(s_t, a_t) \leftarrow \alpha \cdot \hat{Q}(s_t, a_t) + (1-\alpha) Q(s_t, a_t)$$

“better Q(s,a)”

Q-learning VS SARSA

Update rule (from Bellman eq.)

$$Q(s_t, a_t) \leftarrow \alpha \cdot \hat{Q}(s_t, a_t) + (1-\alpha) Q(s_t, a_t)$$

“better Q(s,a)”

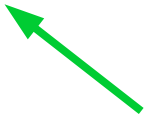
Q-learning

$$\hat{Q}(s, a) = r(s, a) + \gamma \cdot \max_{a'} Q(s', a')$$

Q-learning VS SARSA

Update rule (from Bellman eq.)

$$Q(s_t,a_t) \leftarrow \alpha \cdot \hat{Q}(s_t,a_t) + (1-\alpha) Q(s_t,a_t)$$



“better Q(s,a)”

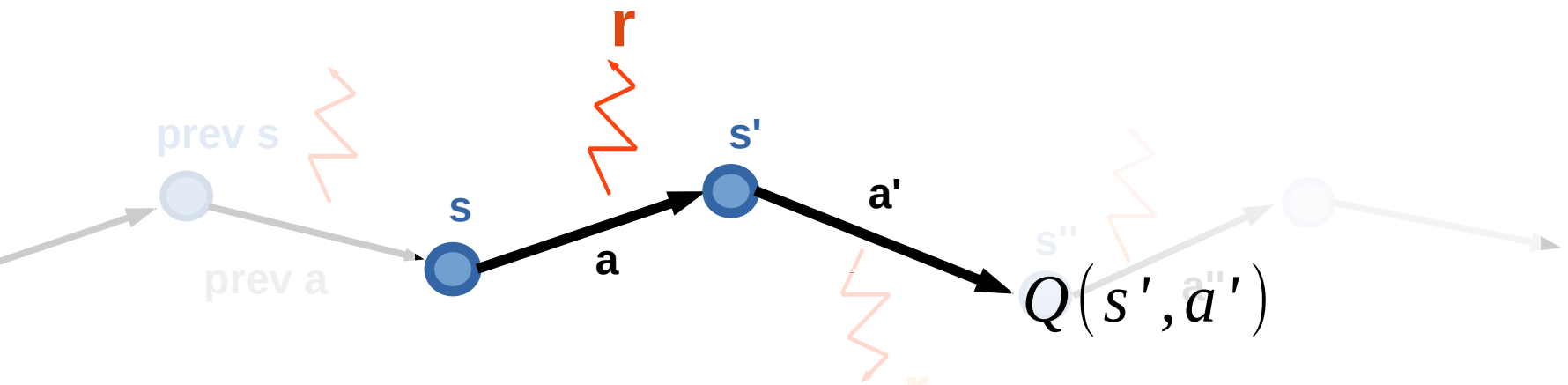
Q-learning

$$\hat{Q}(s,a) = r(s,a) + \gamma \cdot \max_{a'} Q(s',a')$$

SARSA

$$\hat{Q}(s,a) = r(s,a) + \gamma \cdot \underset{a' \sim \pi(a'|s')}{E} Q(s',a')$$

SARSA

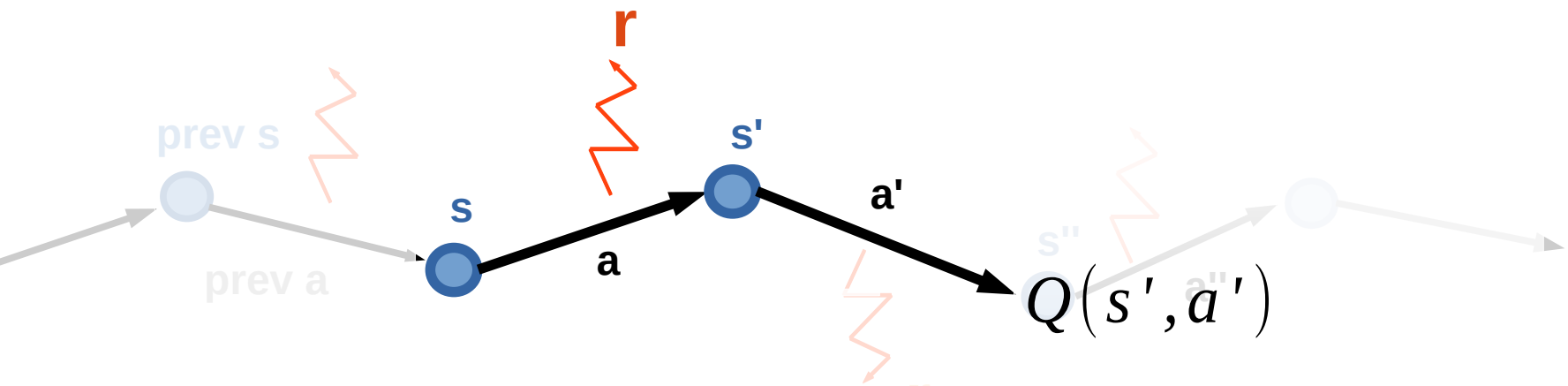


$$\forall s \in S, \forall a \in A, Q(s,a) \leftarrow 0$$

Loop:

- Sample <**s**,**a**,**r**,**s'**,**a'**> from env
- Compute $\hat{Q}(s,a) = r(s,a) + \gamma Q(s',a')$
- Update $Q(s,a) \leftarrow \alpha \cdot \hat{Q}(s,a) + (1-\alpha) Q(s,a)$

SARSA



$$\forall s \in S, \forall a \in A, Q(s,a) \leftarrow 0$$

Loop:

- Sample <**s**,**a**,**r**,**s'**,**a'**> from env

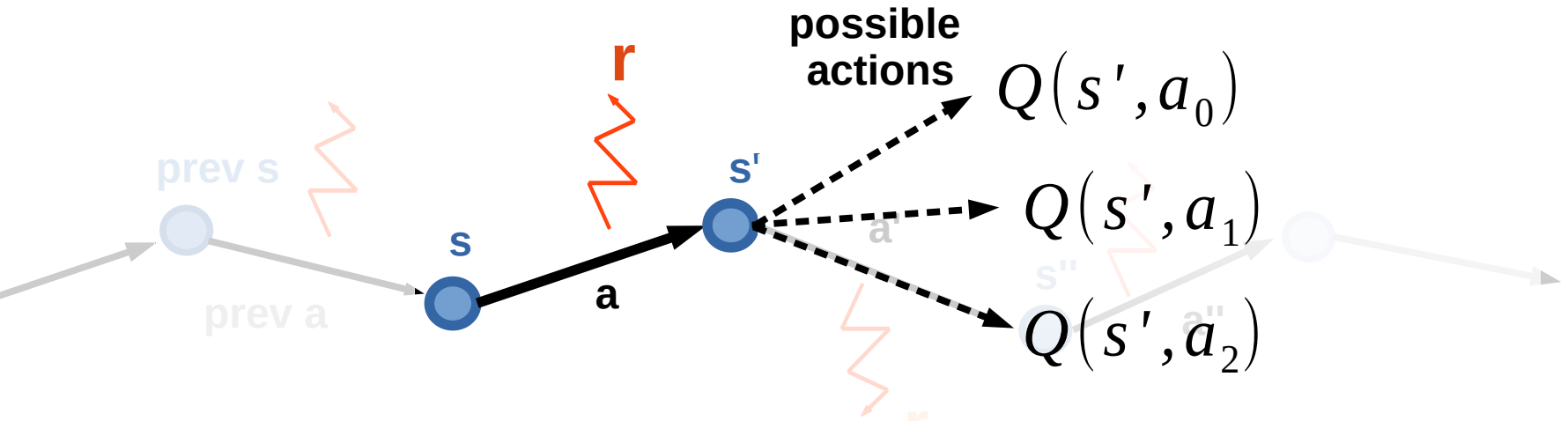
hence "SARSA"

- Compute $\hat{Q}(s,a) = r(s,a) + \gamma Q(s',a')$

next action (not max)

- Update $Q(s,a) \leftarrow \alpha \cdot \hat{Q}(s,a) + (1-\alpha) Q(s,a)$

Expected value SARSA

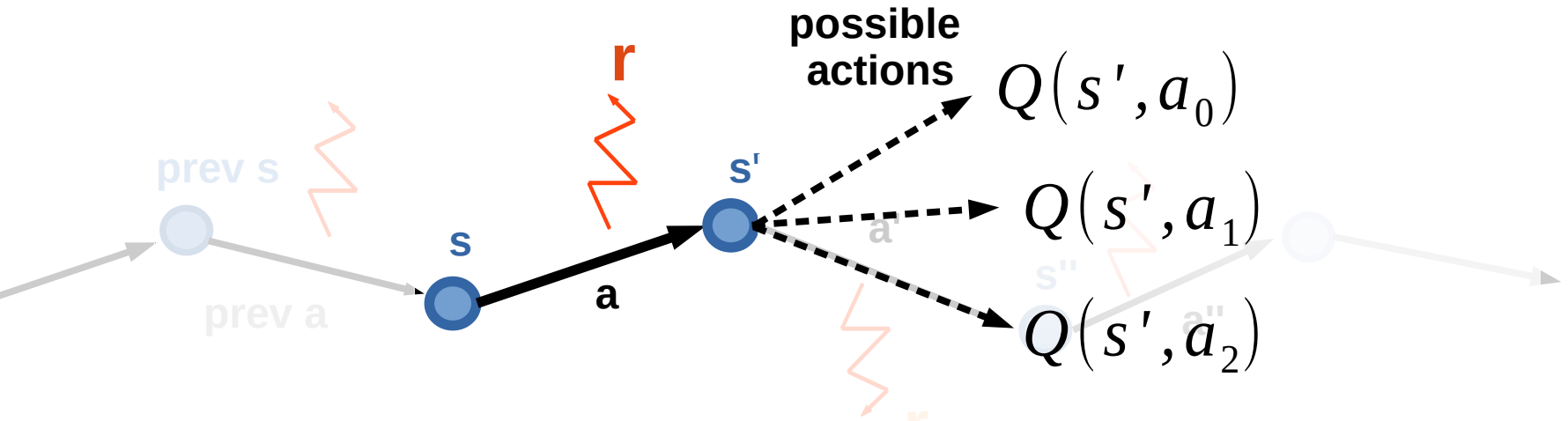


$$\forall s \in S, \forall a \in A, Q(s,a) \leftarrow 0$$

Loop:

- Sample <**s**,**a**,**r**,**s'**> from env
- Compute $\hat{Q}(s,a) = r(s,a) + \gamma \underset{a_i \sim \pi(a|s')}{E} Q(s',a_i)$
- Update $Q(s,a) \leftarrow \alpha \cdot \hat{Q}(s,a) + (1-\alpha) Q(s,a)$

Expected value SARSA



$$\forall s \in S, \forall a \in A, Q(s,a) \leftarrow 0$$

Loop:

- Sample <**s**,**a**,**r**,**s'**> from env

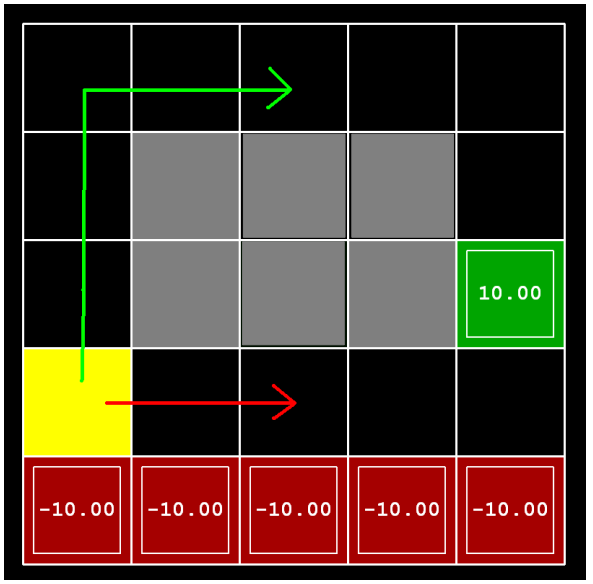
Expected value

- Compute $\hat{Q}(s,a) = r(s,a) + \gamma \underset{a_i \sim \pi(a|s')}{E} Q(s',a_i)$

- Update $Q(s,a) \leftarrow \alpha \cdot \hat{Q}(s,a) + (1-\alpha) Q(s,a)$

Difference

- SARSA gets optimal rewards under current policy
- Q-learning policy **would be** optimal under

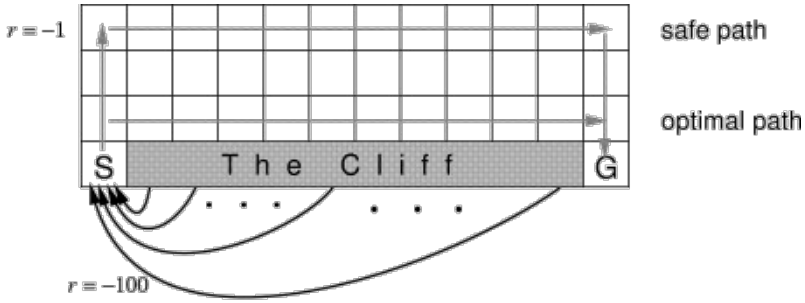


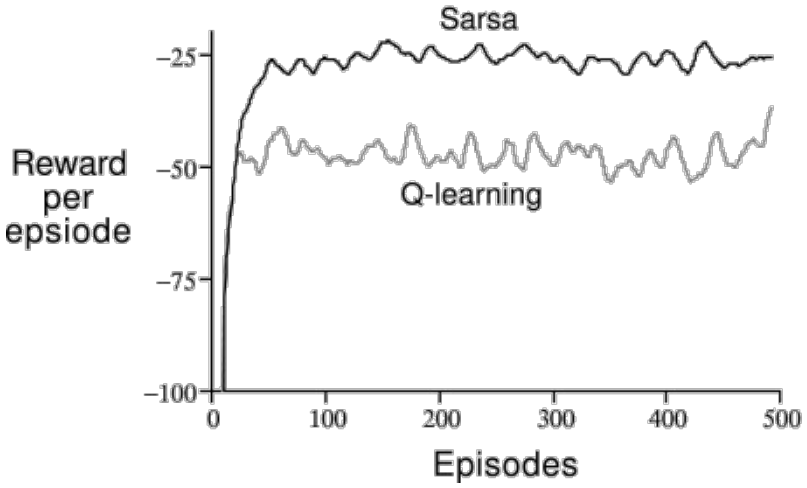
Q-learning

SARSA

Difference

- SARSA converges to optimal policy
- Q-learning policy **would be** optimal without exploration





On-policy vs Off-policy

Two problem setups

on-policy

Agent **can** pick actions

- Most obvious setup :)
- Agent always follows his **own** policy

off-policy

Agent **can't** pick actions

- Learning with exploration, playing without exploration
- Learning from expert (expert is imperfect)
- Learning from sessions (recorded data)

On-policy vs Off-policy

Two problem setups

on-policy

Agent **can** pick actions

- On-policy algorithms **can't** learn off-policy

off-policy

Agent **can't** pick actions

- Off-policy algorithms **can** learn on-policy

learn optimal policy even if agent takes random actions

Q: which of Q-learning, SARSA and exp. val. SARSA will **only** work on-policy?

On-policy vs Off-policy

Two problem setups

on-policy

Agent **can** pick actions

- On-policy algorithms **can't** learn off-policy
- SARSA
- more later

off-policy

Agent **can't** pick actions

- Off-policy algorithms **can** learn on-policy
- Q-learning
- Expected Value SARSA

On-policy vs Off-policy

Two problem setups

on-policy

Agent **can** pick actions

- On-policy algorithms **can't** learn off-policy
- SARSA
- more coming soon

off-policy

Agent **can't** pick actions

- Off-policy algorithms **can** learn on-policy
- Q-learning
- Expected Value SARSA

On-policy vs Off-policy

Two problem setups

on-policy

Agent **can** pick actions

- On-policy algorithms **can't** learn off-policy
- SARSA
- more coming soon

off-policy

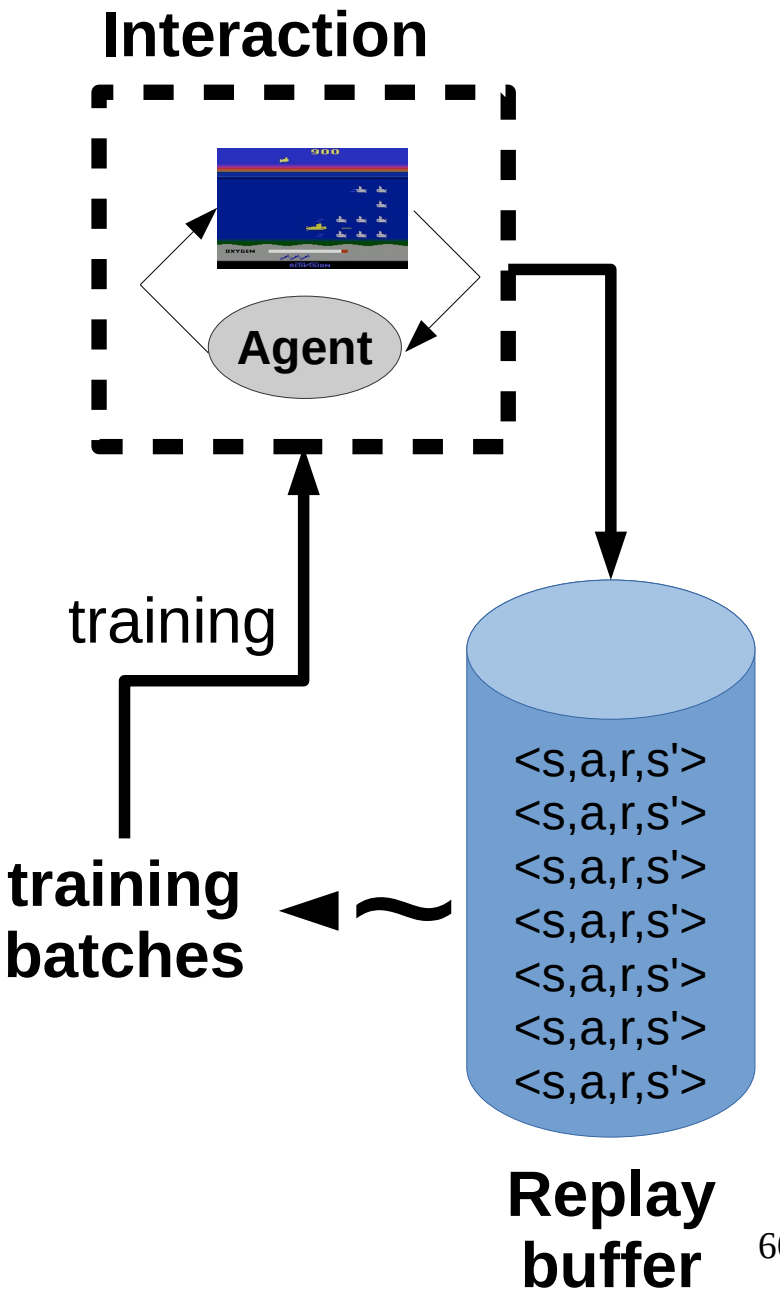
Agent **can't** pick actions

- Off-policy algorithms **can** learn on-policy
- Q-learning
- Expected Value SARSA

Experience replay

Idea: store several past interactions <*s,a,r,s'*>

Train on random subsamples

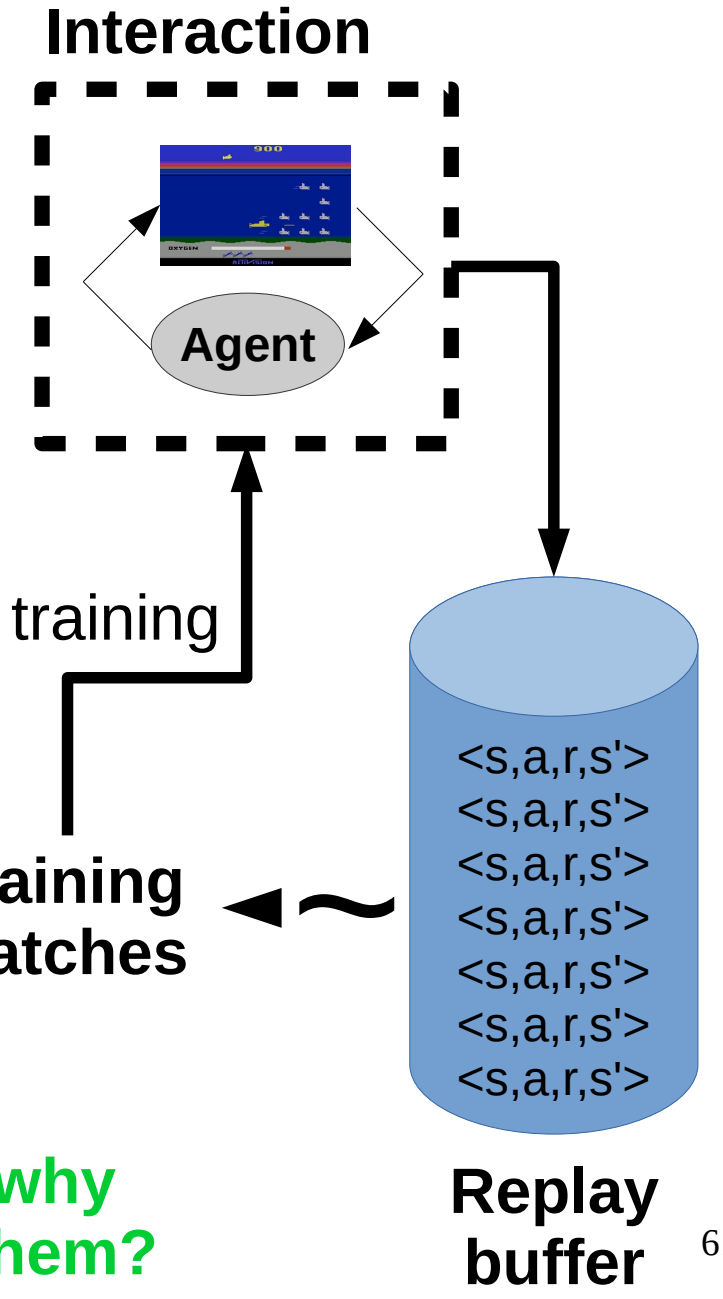


Experience replay

Idea: store several past interactions <*s,a,r,s'*>
Train on random subsamples

Training curriculum:
- play 1 step and record it
- pick N random transitions to train

Profit: you don't need to re-visit same (s,a) many times to learn it.



Only works with off-policy algorithms!

Btw, why only them?

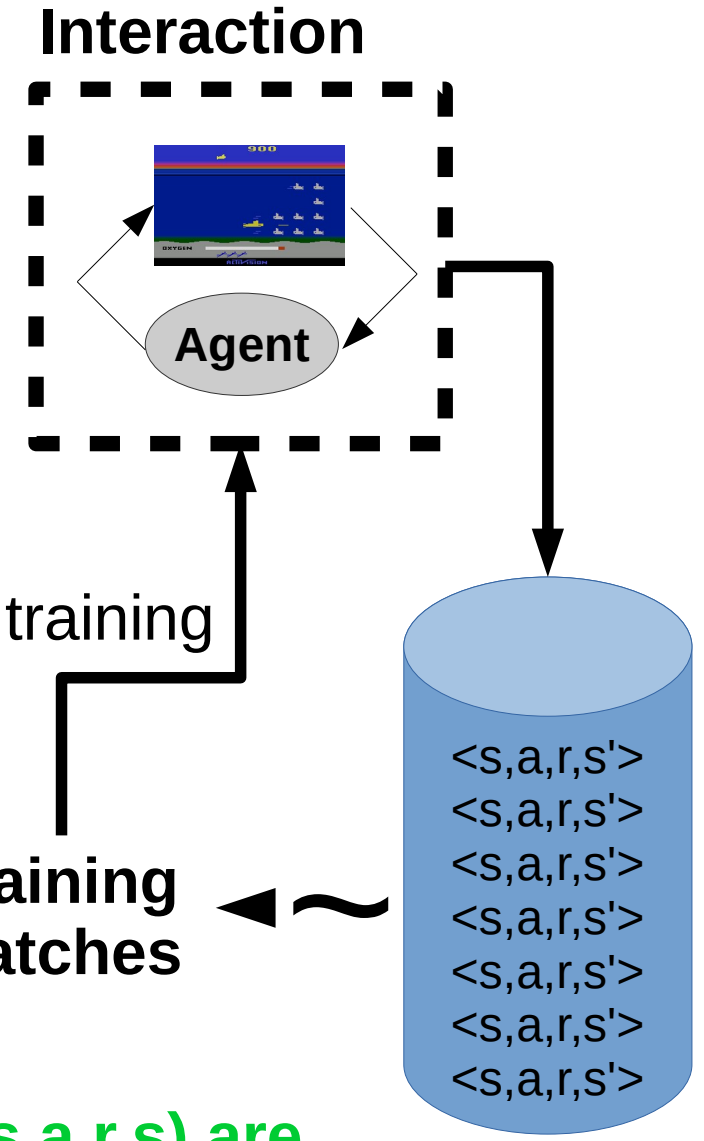
</chapter>

Experience replay

Idea: store several past interactions <*s,a,r,s'*>
Train on random subsamples

Training curriculum:
- play 1 step and record it
- pick N random transitions to train

Profit: you don't need to re-visit same (s,a) many times to learn it.



Only works with off-policy algorithms!

Old (s,a,r,s) are from older/weaker version of policy!

N-step algorithms

Recall R?

$$R_t = r_t + \gamma \cdot r_{t+1} + \gamma^2 \cdot r_{t+2} + \ldots =$$
$$= r_t + \gamma \cdot \left(r_{t+1} + \gamma \cdot r_{t+2} + \ldots \right) =$$
$$= r_t + \gamma \cdot R_{t+1} =$$
$$= r_t + \gamma \cdot r_{t+1} + \gamma^2 \cdot R_{t+2}$$

N-step SARSA

- General formula

$$Q(s_t, a_t) \leftarrow \alpha \cdot \hat{Q}(s_t, a_t) + (1-\alpha) Q(s_t, a_t)$$

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma Q(s_{t+1}, a_{t+1})$$

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma r(s_{t+1}, a_{t+1}) + \gamma^2 Q(s_{t+2}, a_{t+2})$$

N-step SARSA

- General formula

$$Q(s_t, a_t) \leftarrow \alpha \cdot \hat{Q}(s_t, a_t) + (1-\alpha) Q(s_t, a_t)$$

1-step SARSA

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma Q(s_{t+1}, a_{t+1})$$

2-step SARSA

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma r(s_{t+1}, a_{t+1}) + \gamma^2 Q(s_{t+2}, a_{t+2})$$

3-step SARSA

$$\hat{Q}(s_t, a_t) = \text{???}$$

N-step SARSA

- General formula

$$Q(s_t, a_t) \leftarrow \alpha \cdot \hat{Q}(s_t, a_t) + (1-\alpha) Q(s_t, a_t)$$

1-step SARSA

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma Q(s_{t+1}, a_{t+1})$$

2-step SARSA

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma r(s_{t+1}, a_{t+1}) + \gamma^2 Q(s_{t+2}, a_{t+2})$$

3-step SARSA

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma r(s_{t+1}, a_{t+1}) + \gamma^2 r(s_{t+2}, a_{t+2}) + \gamma^3 Q(s_{t+3}, a_{t+3})$$

N-step SARSA

- General formula

$$Q(s_t, a_t) \leftarrow \alpha \cdot \hat{Q}(s_t, a_t) + (1-\alpha) Q(s_t, a_t)$$

1-step SARSA

$$\hat{Q}(s_t, a_t) = r(s_t, a_t) + \gamma Q(s_{t+1}, a_{t+1})$$

N-step SARSA

$$\hat{Q}(s_t, a_t) = [\sum_{\tau=t}^{\tau<t+n} \gamma^{\tau} r(s_{t+\tau}, a_{t+\tau})] + \gamma^{n} Q(s_{t+n}, a_{t+n})$$

N-step algorithms

- General formula

$$Q(s_t,a_t) \leftarrow \alpha \cdot \hat{Q}(s_t,a_t) + (1-\alpha) Q(s_t,a_t)$$

N-step SARSA

$$\hat{Q}(s_t,a_t) = [\sum_{\tau=t}^{\tau<t+n} \gamma^{\tau} r(s_{t+\tau},a_{t+\tau})] + \gamma^{n} Q(s_{t+n},a_{t+n})$$

N-step Q-learning

$$\hat{Q}(s_t,a_t) = [\sum_{\tau=t}^{\tau<t+n} \gamma^{\tau} r(s_{t+\tau},a_{t+\tau})] + \gamma^{n} \cdot \max_{a} Q(s_{t+n},a)$$

Trivia: which of these methods work off-policy?

N-step algorithms

- General formula

$$Q(s_t,a_t) \leftarrow \alpha \cdot \hat{Q}(s_t,a_t) + (1-\alpha) Q(s_t,a_t)$$

N-step SARSA

$$\hat{Q}(s_t,a_t) = [\sum_{\tau=t}^{\tau<t+n} \gamma^{\tau} r(s_{t+\tau},a_{t+\tau})] + \gamma^{n} Q(s_{t+n},a_{t+n})$$

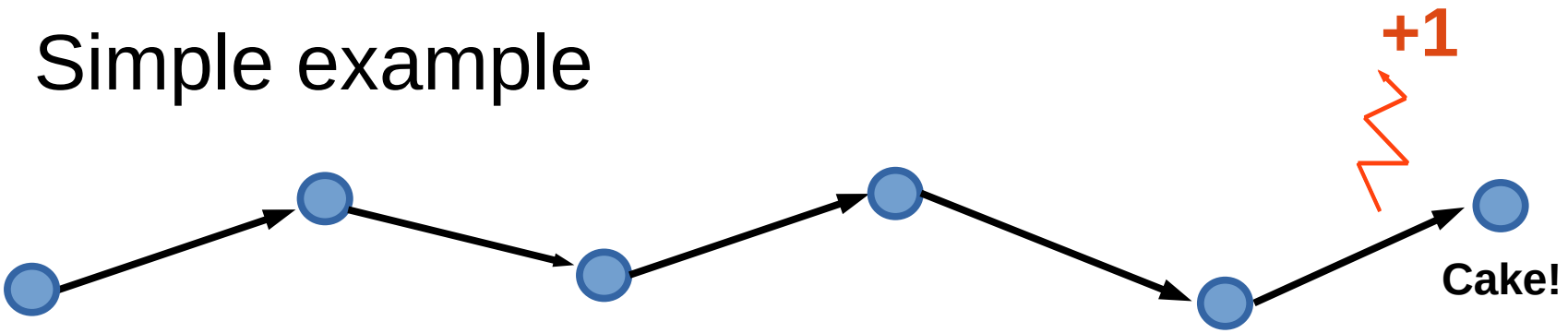
N-step Q-learning

$$\hat{Q}(s_t,a_t) = [\sum_{\tau=t}^{\tau<t+n} \gamma^{\tau} r(s_{t+\tau},a_{t+\tau})] + \gamma^{n} \cdot \max_{a} Q(s_{t+n},a)$$

Trivia: which of these methods work off-policy? **None of them!**

1-step Vs n-step

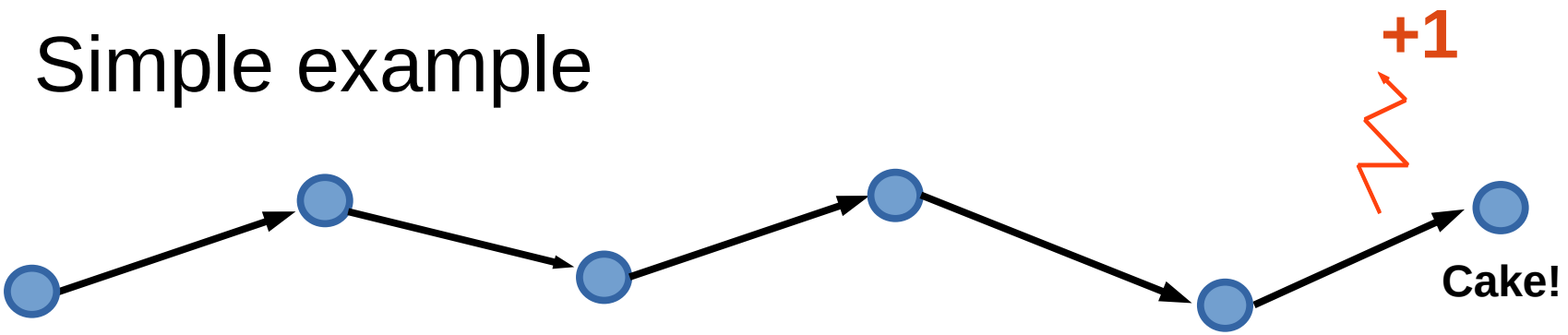
- Simple example



How many games does it take for **SARSA** to learn the optimal policy?

1-step Vs n-step

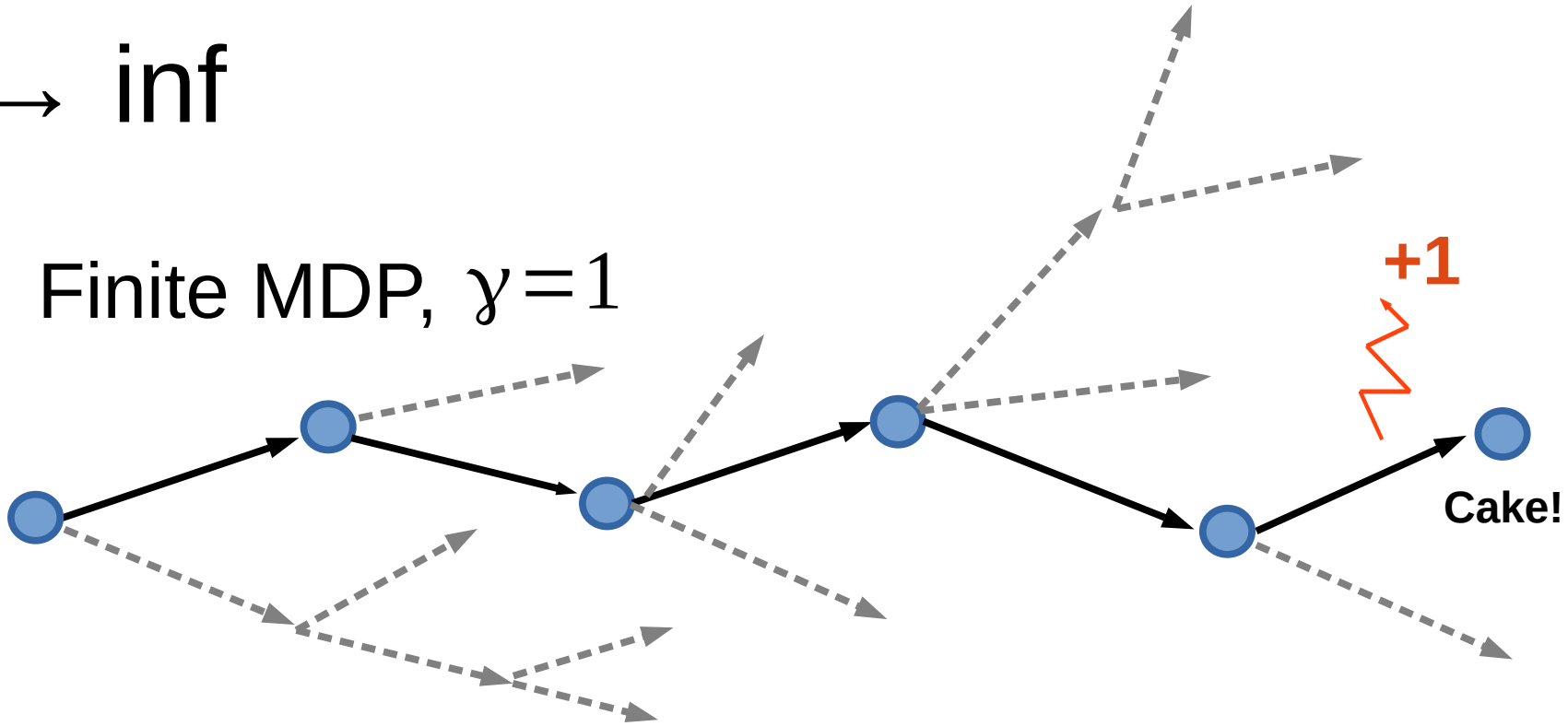
- Simple example



- SARSA needs 5 steps, n-step SARSA needs 1
- Nuts and bolts
  - Nonlinear approximations learn much faster!
  - Play for N steps, than learn (batched)

n → inf

- Finite MDP, $\gamma = 1$



- Sample many trajectories (or tree search)
- Compute expected $Q(s,a) = \underset{\substack{s' \sim p(s'|s,a), \\ a' \sim \pi(a'|s') \\ s'' \sim p(s''|s',a') \\ \dots}}{E} R(s,a)$

minimal assumptions, unbiased, large variance

New stuff we learned

- Anything?

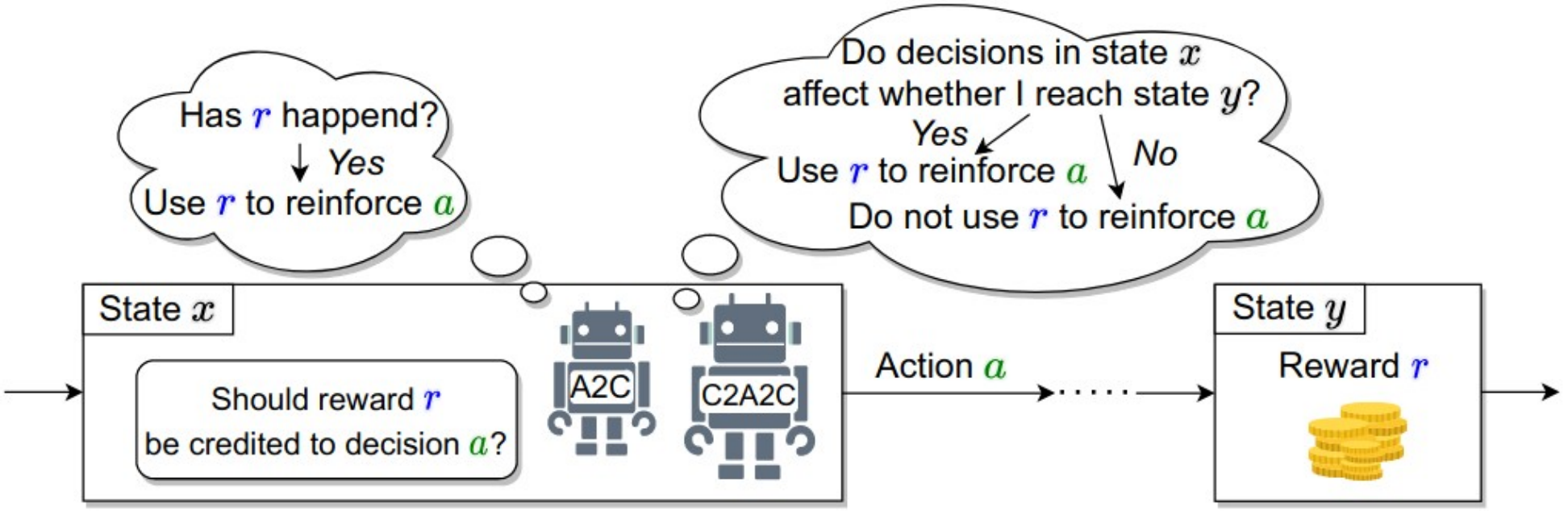
New stuff we learned

- Q(s,a),Q*(s,a)
- Q-learning, SARSA
  - We can learn from trajectories (model-free)
- Exploration vs exploitation (basics)
- Learning On-policy vs Off-policy
  - Using experience replay

Bonus: credit assignment

https://arxiv.org/abs/2011.09464

https://arxiv.org/abs/2106.04499



Try to learn whether (or not) action ***a*** actually caused a future reward **r**

Coming next...

- What if state space is large/continuous
  - Deep reinforcement learning